

Complete analytical solution to the Cornell potential and heavy quarkonium structure

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We use the recently proposed supersymmetric expansion algorithm to obtain a complete analytical solution to the Schrödinger equation with the Cornell potential. We find that the energy levels $E_{nl}(\lambda)$ depend on n^2 and $L^2 = l(l+1)$. For a given n , the energy decreases with l and the radial probabilities have the Coulomb shape but their peaks are shifted toward smaller radius. We study the heavy quarkonium structure on the light of these results, showing that the measured $\bar{b}b$ and $\bar{c}c$ meson masses follow the inverted spectrum pattern predicted by the Cornell potential. Details of the structure of heavy quarkonium like the mean inverse radius and mean squared velocity for the different quarkonium configurations can be obtained from our solution. These details point to significant relativistic corrections for all the configurations of real heavy quarkonium. We calculate relativistic corrections using perturbation theory finding an expansion in α_s^2 for the heavy quarkonium masses. The mass hierarchies in the fine splittings can be qualitatively understood from this expansion. The quantitative analysis of the Bohr-like levels and of the fine splittings in the $l=0$ sector allow us to make well-defined predictions for the masses of some of the missing heavy quarkonium states, to identify the $\psi(4040)$ as the $3^3S_1 \bar{c}c$ state and the $\psi(3842)$, $\psi(3823)$ and $\psi(3770)$ as the 3^3D_3 , 3^3D_2 , and $3^3D_1 \bar{c}c$ states, respectively.

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I. INTRODUCTION

The Cornell potential, also named Coulomb plus linear or funnel potential in the literature, was one of the first models for the phenomenological description of the confined dynamics of quarkonium [1–3]. The potential is a sum of a Coulomb term and a linear term. The Coulomb term is motivated from the perturbative short-distance regime of QCD. At intermediate and long distances (compared to the typical size of the heavy quarkonium), we face the unsolved problem of confinement.

The formal derivation of quark-antiquark forces from QCD started with the pioneering work by Wilson [4] who related the interquark potential to the so-called Wilson loop. The evaluation of the Wilson loop in the long-distance regime, with diverse methods, yields a linear

potential as the leading term [5–12]. The physical picture that arises at long distances under reasonable assumptions is an interquark potential generated by a chromoelectric flux tube [8] which produces a QCD string. These results put on a firm basis the Coulomb plus linear potential as a suitable starting point for the phenomenological description of the structure of heavy quarkonium.

The systematic calculation of heavy quarkonium properties from QCD has had considerable advances with the formulation of effective field theories for QCD in the nonrelativistic regime. The nonrelativistic expansion of QCD (NRQCD) [13] and the potential nonrelativistic QCD expansion (pNRQCD) [14], take care of the scales of heavy quarkonium, integrate physics at high energy and yield systematic expansions in terms of operators suppressed according to the corresponding power counting rules. In the case of NRQCD there are two scales, m_Q and $m_Q v^2$, with a well-defined hierarchy $m_Q \gg m_Q v^2$, and the effective theory is obtained integrating out the degrees of freedom at the scale m_Q . For heavy quarkonium, a hierarchy $m_Q \gg m_Q v \gg m_Q v^2 \gg \Lambda_{\text{QCD}}$ is expected and pNRQCD obtains the effective theory for physics at the

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ultrasoft scale $m_Q v^2$ integrating out also physics at the soft scale $m_Q v$. A primary concern for the p NRQCD is the validity of this scale hierarchy which grants the validity of the perturbative matching which otherwise must be done considering the strong coupling regime. Presently, it is not clear if all heavy quarkonium configurations respect the above hierarchy (the weak coupling scenario) or for some states actually it is necessary to work in the strong coupling matching regime. On the other side, calculations in the lattice became efficient with the use of effective field theories and yield the same conclusion; the nonrelativistic description of QCD in the nonperturbative region is given by the Cornell potential [15,16]. A complete analytical solution to the Cornell potential is desirable because it can shed light into the structure of real heavy quarkonium, especially of the relative size of the soft and ultrasoft scales and Λ_{QCD} .

The approach followed in Refs. [1–3] and almost in every subsequent work on analytical solutions to the Cornell potential, was to consider first the linear potential whose exact solutions for S -waves are given in terms of the Airy functions, and to incorporate later the effects of the Coulomb part using Rayleigh-Schrödinger perturbation theory to obtain approximate solutions with an expansion in powers of the strong coupling constant α_s . The lack of complete analytical solutions to the Cornell potential motivated the use of general results aiming to guess from nonrelativistic quantum mechanics the structure of heavy quarkonium [17,18], but most of the work on the field has been done using numerical solutions for the Cornell potential [19–30].

Although such solutions have been used for almost half a century to get insight on the heavy quarkonium dynamics, the relative importance of the nonperturbative effects modeled by the linear term and the perturbative effects has not been clearly established, precisely because the latter are incorporated only in a perturbative manner. Recently, there has been renewed interest in the heavy quarkonium spectrum due to the possibilities opened by new factories for the discovery of conventional heavy quarkonium as well as exotic states (for a review on these possibilities and a more complete list of references to old literature see [31–33]).

A review and classification of the different methods aiming to obtain analytical solutions to the Schrödinger equation and a novel systematic method named the supersymmetric expansion algorithm (SEA) was recently published [34]. The SEA allows us to use the full power of supersymmetry in nonrelativistic quantum mechanics and complete analytical solutions to longstanding unsolved potentials relevant to several branches of natural sciences like the Yukawa potential [35,36], Hulthén and anharmonic potentials [34] have been obtained by this method. In this work, we use the SEA to obtain the complete analytical solutions of the Cornell potential, apply the solutions to the description of the physics of heavy quarkonium and

explore the leading relativistic corrections. We are able to extract general qualitative and quantitative predictions for the bottomonium and charmonium spectrum which nicely agree with existing data.

This paper is organized as follows. In the next section we give the detail of the calculation of the complete analytical solution for the Cornell potential. Section III is devoted to the calculation of the heavy quarkonium spectrum using the solutions to the Cornell potential and its comparison with existing data. In Sec. IV we consider the leading relativistic corrections that yield the fine splittings. Section V is devoted to a first quantitative analysis of the fine splittings in the $l = 0$ sector, qualitative analysis of the fine splittings for $l = 1, 2$ and the corresponding quantitative and qualitative predictions for the bottomonium and charmonium spectrum. We close with our conclusions and perspectives in Sec. VI.

II. THE CORNELL POTENTIAL IN THE SEA

The Cornell potential is given by

$$V_c(r) = -\frac{\alpha \hbar c}{r} + \frac{\sigma}{\hbar c} r. \quad (1)$$

We introduce in this section the $\hbar$ and c factor in order to have a dimensionless parameter α and a parameter σ with units of E^2 . In the remaining of this section we will solve the Schrödinger equation (SE) for the Coulomb plus linear potential using the supersymmetric expansion algorithm introduced in Ref. [34]. We refer the reader to that work for further details of the formalism. In a first step, we use the typical distance scale of the system as the scale related to the coupling α , i.e., the Bohr radius

$$a = \hbar / \mu \alpha, \quad (2)$$

where μ is the reduced mass of the system, to cast the radial part of the Schrödinger equation into its dimensionless form,

$$\left[-\frac{d^2}{dx^2} + v_0(x, \lambda) \right] u_0(x, \lambda) = \epsilon_0(\lambda) u_0(x, \lambda), \quad (3)$$

where a suffix “0” is attached to all quantities because this is the initial step of the algorithm. Here, $x = r/a$, $R(r) = u_0(x)/x$, $\epsilon_0(\lambda) = E_0(\lambda)/E_c$, with the typical energy scale given by the Coulomb energy $E_c = \hbar^2 / 2\mu a^2 = \mu c^2 \alpha^2 / 2$. The dimensionless effective potential for the Cornell potential is given by

$$v_0(x, \lambda) = \frac{l(l+1)}{x^2} - \frac{2}{x} + \lambda x, \quad (4)$$

with the normalized string tension

$$\lambda = \frac{\sigma}{E_c E_s} = \frac{2\sigma}{(\mu c^2)^2 \alpha^3}, \quad (5)$$

where $E_s = \hbar c/a = \mu c^2 \alpha$.

The SEA constructs the complete analytical solution starting with very special states named edge states, which in general are nodeless (excited or ground) states. With this aim, we first recast the SE into a logarithmic form,

$$W_0^2(x, \lambda) - W_0' = v_0(x, \lambda) - \epsilon_0(\lambda), \quad (6)$$

where

$$W_0(x, \lambda) = -\frac{d}{dx} \ln u_0(x, \lambda). \quad (7)$$

The solution to Eq. (6) is obtained as power series in λ ,

$$\begin{aligned} v_0(x, \lambda) &= \sum_{k=0}^{\infty} v_{0k}(x) \lambda^k, \\ W_0(x, \lambda) &= \sum_{k=0}^{\infty} w_{0k}(x) \lambda^k, \quad \epsilon_0(\lambda) = \sum_{k=0}^{\infty} \epsilon_{0k} \lambda^k. \end{aligned} \quad (8)$$

The coefficients for the expansion of the Cornell potential in powers of λ are given by

$$v_{0k}(x) = \begin{cases} \frac{l(l+1)}{x^2} - \frac{2}{x} & \text{for } k=0, \\ x & \text{for } k=1, \\ 0 & \text{for } k \geq 2. \end{cases} \quad (9)$$

The coefficients in the power series of $W_0(x, \lambda)$ and $\epsilon_0(\lambda)$ satisfy the following infinite set of hierarchical equations,

$$k=0: w_{00}^2 - w_{00}' = \frac{l(l+1)}{x^2} - \frac{2}{x} - \epsilon_{00}, \quad (10)$$

$$k=1: 2w_{00}w_{01} - w_{01}' = x - \epsilon_{01}, \quad (11)$$

$$k \geq 2: 2w_{00}w_{0k} - w_{0k}' = -B_{0k}(x) - \epsilon_{0k}, \quad (12)$$

where

$$B_{0k} = \sum_{m+n=k} w_{0m} w_{0n}. \quad (13)$$

The solutions up to $k=5$ are given by

$$w_{00} = \frac{1}{b} - \frac{b}{x}, \quad (14)$$

$$w_{01} = \frac{b}{2}x, \quad (15)$$

$$w_{02} = -\frac{b^3}{8}[b(b+1)x + x^2], \quad (16)$$

$$w_{03} = \frac{b^5}{32}[b^2(b+1)(4b+5)x + b(4b+5)x^2 + 2x^3], \quad (17)$$

$$\begin{aligned} w_{04} = & -\frac{b^7}{256}[b^3(b+1)(48b^2 + 129b + 88)x \\ & + b^2(48b^2 + 129b + 88)x^2 \\ & + 2b(15b + 22)x^3 + 10x^4], \end{aligned} \quad (18)$$

$$\begin{aligned} w_{05} = & \frac{b^9}{512}[b^4(b+1)(176b^3 + 753b^2 + 1049b + 539)x \\ & + b^3(176b^3 + 753b^2 + 1049b + 539)x^2 \\ & + 2b^2(60b^2 + 185b + 147)x^3 \\ & + b(56b + 93)x^4 + 14x^5], \end{aligned} \quad (19)$$

$$\epsilon_{00} = -\frac{1}{b^2}, \quad (20)$$

$$\epsilon_{01} = \frac{b}{2}(2b+1), \quad (21)$$

$$\epsilon_{02} = -\frac{b^4}{8}(b+1)(2b+1), \quad (22)$$

$$\epsilon_{03} = \frac{b^7}{32}(b+1)(2b+1)(4b+5), \quad (23)$$

$$\epsilon_{04} = -\frac{b^{10}}{256}(b+1)(2b+1)(48b^2 + 129b + 88), \quad (24)$$

$$\epsilon_{05} = \frac{b^{13}}{512}(b+1)(2b+1)(176b^3 + 753b^2 + 1049b + 539), \quad (25)$$

where

$$b = l + 1. \quad (26)$$

In general, for $k \geq 2$ we have polynomial solutions for $w_{0k}(x)$,

$$w_{0k}(x) = \sum_{\alpha=1}^k w_{0k\alpha} x^\alpha, \quad (27)$$

and inserting this expression in Eq. (12) we get the following recursion relations for the constant coefficients $w_{0k\alpha}$:

$$w_{0kk} = -\frac{b}{2}B_{0kk}, \quad (28)$$

$$\begin{aligned} w_{0k\alpha} = & \frac{b}{2}((2b + \alpha + 1)w_{0k(\alpha+1)} - B_{0k\alpha}), \\ & \alpha = k-1, k-2, \dots, 3, 2, \end{aligned} \quad (29)$$

$$\epsilon_{0k} = (2b+1)w_{0k1}, \quad (30)$$

where

$$B_{0k\alpha} = \sum_{m+n=k} \sum_{\beta+\gamma=\alpha} w_{0m\beta} w_{0n\gamma}. \quad (31)$$

These recurrence relations yield the first solution for the Cornell Hamiltonian,

$$H_0 = -\frac{d^2}{dx^2} + v_0(x, \lambda). \quad (32)$$

Indeed, separating the λ -independent terms,

$$W_0(x, \lambda) = \frac{1}{l+1} - \frac{l+1}{x} + \sum_{k=1}^{\infty} w_{0k}(x) \lambda^k, \quad (33)$$

we can see that the λ -independent terms correspond to the solution to the Coulomb problem, thus, for this solution, $l+1 = n$ and skipping normalization factors we get

$$u_0(x, \lambda) = x^n e^{-\frac{x}{n}} e^{-G_0(x, \lambda)} \equiv \phi_{n, n-1}^{(0)}(x, \lambda), \quad (34)$$

$$\epsilon_0(\lambda) = -\frac{1}{n^2} + \sum_{k=1}^{\infty} (2n+1) w_{0k1} \lambda^k, \quad (35)$$

where we used $\phi_{nl}^{(0)}(x, \lambda)$ for the solutions with the principal quantum number n and angular momentum l of the Cornell Hamiltonian H_0 . The function G_0 is given by

$$G_0(x, \lambda) = \sum_{k=1}^{\infty} \lambda^k \left(\sum_{\alpha=1}^k w_{0k\alpha} \frac{x^{\alpha+1}}{\alpha+1} \right). \quad (36)$$

The solution $\phi_{n, n-1}^{(0)}(x, \lambda)$ is the edge state for the n th level of the Cornell Hamiltonian H_0 . The rest of the states in this level are constructed with the aid of supersymmetry. We start noticing that H_0 can be factorized as

$$H_0 = a_0^\dagger a_0 + \epsilon_0, \quad (37)$$

where

$$a_0 = \frac{d}{dx} + W_0, \quad a_0^\dagger = -\frac{d}{dx} + W_0, \quad (38)$$

and the edge state, u_0 , satisfies

$$a_0 u_0(x, \lambda) = 0. \quad (39)$$

Then we construct the supersymmetric partner,

$$H_1 = a_0 a_0^\dagger + \epsilon_0 = -\frac{d^2}{dx^2} + v_1(x, \lambda), \quad (40)$$

with

$$\begin{aligned} v_1(x, \lambda) &= v_0(x, \lambda) + 2W'_0 \\ &= \frac{l(l+1)}{x^2} - \frac{2}{x} + \lambda x + 2 \sum_{k=0}^{\infty} w'_{0k}(x) \lambda^k. \end{aligned} \quad (41)$$

Now we find the edge state solution for H_1 in a similar way as we did it for H_0 , i.e., we recast the Schrödinger equation for H_1 in the logarithmic form,

$$W_1^2 - W'_1 = v_1(x, \lambda) - \epsilon_1(\lambda) = v_0(x, \lambda) + 2W'_0 - \epsilon_1(\lambda), \quad (42)$$

where

$$W_1(x, \lambda) = -\frac{d}{dx} \ln u_1(x, \lambda). \quad (43)$$

The solution to Eq. (42) is obtained as power series in λ ,

$$\begin{aligned} v_1(x, \lambda) &= \sum_{k=0}^{\infty} v_{1k}(x) \lambda^k, & W_1(x, \lambda) &= \sum_{k=1}^{\infty} w_{1k}(x) \lambda^k, \\ \epsilon_1(\lambda) &= \sum_{k=1}^{\infty} \epsilon_{1k} \lambda^k. \end{aligned} \quad (44)$$

The coefficients for the expansion of v_1 in powers of λ are given by

$$v_{1k}(x) = \begin{cases} \frac{l(l+1)}{x^2} - \frac{2}{x} + 2w'_{00} & \text{for } k=0, \\ x + 2w'_{01} & \text{for } k=1, \\ 2w'_{0k} & \text{for } k \geq 2. \end{cases} \quad (45)$$

The coefficients in the power series of $W_1(x, \lambda)$ and $\epsilon_1(\lambda)$ satisfy now the following infinite set of hierarchical equations:

$$k=0: w_{10}^2 - w'_{10} = \frac{(l+1)(l+2)}{x^2} - \frac{2}{x} - \epsilon_{10}, \quad (46)$$

$$k=1: 2w_{10}w_{11} - w'_{11} = x + 2w'_{01} - \epsilon_{11}, \quad (47)$$

$$k \geq 2: 2w_{10}w_{1k} - w'_{1k} = 2w'_{0k} - B_{1k} - \epsilon_{1k}, \quad (48)$$

where

$$B_{1k} = \sum_{m+n=k} w_{1m} w_{1n}. \quad (49)$$

The solutions for $k=0, 1$ are

$$w_{10} = \frac{1}{b+1} - \frac{b+1}{x}, \quad \epsilon_{10} = -\frac{1}{(b+1)^2}, \quad (50)$$

$$w_{11} = \frac{b+1}{2}x, \quad \epsilon_{11} = \frac{1}{2}(3(b+1)^2 - b(b-1)), \quad (51)$$

while for $k \geq 2$, w_{1k} has the general form

$$w_{1k}(x) = \sum_{\alpha=1}^k w_{1k\alpha} x^\alpha, \quad (52)$$

where the numerical coefficients satisfy the following recurrence relations:

$$w_{1kk} = -\frac{b+1}{2} B_{1kk}, \quad (53)$$

$$w_{1k\alpha} = \frac{b+1}{2} ((2b+3+\alpha)w_{0k(\alpha+1)} - B_{1k\alpha}), \quad (54)$$

$$\alpha = k-1, k-2, \dots, 3, 2,$$

$$\epsilon_{1k} = (2b+3)w_{1k1} + 2w_{0k1}. \quad (55)$$

From the λ -independent part of the solution we identify this solution as a $l = n-2$ solution of the level n th of H_1 . Explicitly, the unnormalized solution is given by

$$u_1(x, \lambda) = x^n e^{-\frac{x}{n}} e^{-G_1(x, \lambda)} \equiv \phi_{n, n-2}^{(1)}(x, \lambda), \quad (56)$$

$$\epsilon_1(\lambda) = -\frac{1}{n^2} + \sum_{k=1}^{\infty} ((2b+3)w_{1k1} + 2w_{0k1})\lambda^k. \quad (57)$$

The function G_1 is given by

$$G_1(x, \lambda) = \sum_{k=1}^{\infty} \lambda^k \left(\sum_{\alpha=1}^k w_{1k\alpha} \frac{x^{\alpha+1}}{\alpha+1} \right). \quad (58)$$

Supersymmetry allows us to obtain a second solution to the Cornell Hamiltonian H_0 , from the edge eigenstate u_1 of H_1 . Indeed, u_1 satisfies

$$H_1 u_1(x, \lambda) = [a_0 a_0^\dagger + \epsilon_0] u_1(x, \lambda) = \epsilon_1(\lambda) u_1(x, \lambda). \quad (59)$$

Acting with $a_0^\dagger$ on this equation we get

$$[a_0^\dagger a_0 + \epsilon_0] a_0^\dagger u_1(x, \lambda) = H_0 a_0^\dagger u_1(x, \lambda) = \epsilon_1(\lambda) a_0^\dagger u_1(x, \lambda), \quad (60)$$

thus, $a_0^\dagger u_1(x, \lambda)$ is also an eigenstate of H_0 with eigenvalue $\epsilon_1(\lambda)$. This is also a solution for the n th level but now with $l = n-2$. The unnormalized solution for H_0 is

$$\phi_{n, n-2}^{(0)}(x, \lambda) = a_0^\dagger u_1(x, \lambda) = a_0^\dagger \phi_{n, n-2}^{(1)}(x, \lambda). \quad (61)$$

Notice that in the usual nl notation, $\epsilon_1(\lambda) = \epsilon_{n, n-2}(\lambda)$, $\epsilon_0(\lambda) = \epsilon_{n, n-1}(\lambda)$ and $\epsilon_{n, n-1}(\lambda) \neq \epsilon_{n, n-2}(\lambda)$, thus the l -degeneracy of the n th level of the Coulomb potential is broken by the linear term in the Cornell Hamiltonian H_0 .

We continue this process and construct a new solution to the n th level of H_0 factorizing now H_1 as

$$H_1 = a_1^\dagger a_1 + \epsilon_1(\lambda), \quad (62)$$

with

$$a_1 = -\frac{d}{dx} + W_1(x, \lambda), \quad a_1^\dagger = \frac{d}{dx} + W_1(x, \lambda), \quad (63)$$

and constructing a superpartner to H_1 defined as

$$H_2 = a_1 a_1^\dagger + \epsilon_1(\lambda) = -\frac{d^2}{dx^2} + v_2(x, \lambda), \quad (64)$$

where

$$v_2(x, \lambda) = v_1(x, \lambda) + 2W_1'. \quad (65)$$

We solve likewise this potential obtaining the unnormalized solution of H_0 as

$$\phi_{n, n-3}^{(0)}(x, \lambda) = a_0^\dagger a_1^\dagger u_2(x, \lambda) = a_0^\dagger a_1^\dagger \phi_{n, n-3}^{(2)}(x, \lambda) \quad (66)$$

where $u_2(x, \lambda)$ is the solution to H_2 with eigenvalue $\epsilon_2(\lambda)$ which is a common eigenvalue of H_2 , H_1 , H_0 .

Repeating the algorithm, in the step r we solve the r th superpartner H_r in the same manner. The solutions for $k = 0, 1$ in this step are

$$w_{r0} = \frac{1}{b+r} - \frac{b+r}{x}, \quad \epsilon_{r0} = -\frac{1}{(b+r)^2}, \quad (67)$$

$$w_{r1} = \frac{b+r}{2} x, \quad \epsilon_{r1} = \frac{1}{2} (3(b+r)^2 - b(b-1)), \quad (68)$$

while for $k \geq 2$ the coefficients in the expansion of the r th superpotential W_r has the general form

$$w_{rk}(x) = \sum_{\alpha=1}^k w_{rk\alpha} x^\alpha, \quad (69)$$

where the numerical coefficients satisfy the following recurrence relations:

$$w_{rkk} = -\frac{b+r}{2} B_{rkk}, \quad (70)$$

$$w_{rk\alpha} = \frac{b+r}{2} \left((2b+2r+\alpha+1)w_{0k(\alpha+1)} + 2(\alpha+1) \sum_{q=0}^{r-1} w_{qk(\alpha+1)} - B_{rk\alpha} \right),$$

$$\alpha = k-1, k-2, \dots, 3, 2, \quad (71)$$

$$\varepsilon_{rk} = (2b + 2r + 1)w_{rk1} + 2 \sum_{q=0}^{r-1} w_{qk1}, \quad (72)$$

with

$$B_{rka} = \sum_{m+n=k} \sum_{\beta+\gamma=\alpha} w_{rm\beta} w_{rn\gamma}. \quad (73)$$

The unnormalized solution for the edge state of H_r , is given by

$$u_r(x, \lambda) = x^n e^{-\frac{x}{n}} e^{-G_r(x, \lambda)}, \quad (74)$$

$$\varepsilon_r(\lambda) = -\frac{1}{n^2} + \sum_{k=0}^{\infty} \left((2n+1)w_{rk1} + 2 \sum_{q=0}^{r-1} w_{qk1} \right) \lambda^k, \quad (75)$$

where

$$G_r(x, \lambda) = \int w_r(x, \lambda) dx = \sum_{k=1}^{\infty} \lambda^k \left(\sum_{\alpha=1}^k w_{rk\alpha} \frac{x^{\alpha+1}}{\alpha+1} \right). \quad (76)$$

The solution for the Cornell Hamiltonian H_0 in the nl notation is given by

$$\phi_{n,n-1-r}^{(0)} = a_0^\dagger a_1^\dagger \dots a_{r-2}^\dagger a_{r-1}^\dagger u_r, \quad (77)$$

$$\varepsilon_{n,n-1-r}(\lambda) = -\frac{1}{n^2} + \sum_{k=0}^{\infty} \left((2n+1)w_{rk1} + 2 \sum_{q=0}^{r-1} w_{qk1} \right) \lambda^k. \quad (78)$$

The process terminates for $r = n - 1$ when we reach the lowest value $l = 0$ and the n th level of the normalized Cornell Hamiltonian H_0 is completely solved. Notice that we started with an arbitrary value of n thus, all the levels of

the Cornell potential can be solved with the supersymmetric expansion algorithm. All the information of the analytic solution resides in the coefficients $w_{rk\alpha}$ of the expansion in Eq. (69) which satisfy the algebraic recurrence relations in Eqs. (70)–(72). We wrote a symbolic code to solve these relations to the desired order in λ .

We find that the energy levels $\varepsilon_{nl}(\lambda)$ can be written as

$$\varepsilon_{nl}(\lambda) = \sum_{k=0}^{\infty} \varepsilon_k(n^2, L^2) \lambda^k, \quad (79)$$

where the coefficients ε_k depend on n^2 and $L^2 = l(l+1)$. The coefficients for large k have long expressions thus we explicitly write them only up to $k = 10$ for future reference,

$$\varepsilon_0(n^2, L^2) = -\frac{1}{n^2}, \quad (80)$$

$$\varepsilon_1(n^2, L^2) = \frac{1}{2}(3n^2 - L^2), \quad (81)$$

$$\varepsilon_2(n^2, L^2) = -\frac{n^2}{16}(7n^4 - 3L^4 + 5n^2), \quad (82)$$

$$\varepsilon_3(n^2, L^2) = \frac{n^4}{64}(33n^6 - 7n^2L^4 - 10L^6 + 75n^4), \quad (83)$$

$$\varepsilon_4(n^2, L^2) = -\frac{n^6}{512}(465n^8 - 99n^4L^4 - 90n^2L^6 - 84L^8 + 2275n^6 - 180n^2L^4 + 440n^4), \quad (84)$$

$$\varepsilon_5(n^2, L^2) = \frac{n^8}{1024}(1995n^{10} + 17340n^8 - 3n^6(155L^4 - 3803) - 91n^4L^4(4L^2 + 23) - 88n^2L^6(3L^2 + 10) - 198L^{10}), \quad (85)$$

$$\begin{aligned} \varepsilon_6(n^2, L^2) = & -\frac{n^{10}}{16385}(77027n^{12} + 1060290n^{10} - 133n^8(150L^4 - 11663) \\ & - 34n^6(426L^6 + 4887L^4 - 5000) - 15n^4L^4(607L^4 + 5612L^2 + 4800) \\ & - 364n^2L^8(17L^2 + 75) - 4004L^{12}), \end{aligned} \quad (86)$$

$$\begin{aligned} \varepsilon_7(n^2, L^2) = & \frac{n^{12}}{131072}(1608201n^{14} + 32473350n^{12} - 69n^{10}(6698L^4 - 1254497) \\ & - 66n^8(4858L^6 + 93961L^4 - 485630) - 19n^6L^4(9525L^4 + 175460L^2 + 402116) \\ & - 204n^4L^6(543L^4 + 6378L^2 + 10000) - 24480n^2L^{10}(3L^2 + 16) - 66432L^{14}), \end{aligned} \quad (87)$$

$$\begin{aligned}
\epsilon_8(n^2, L^2) = & -\frac{n^{14}}{2097152} (71016319n^{16} + 1991448850n^{14} - 1323n^{12}(17018L^4 - 6494305) \\
& + 650n^{10}(10847042 - 687771L^4 - 23310L^6) \\
& - 69n^8(112885L^8 + 3576100L^6 + 15774908L^4 - 8000000) \\
& - 924n^6L^4(4637L^6 + 106904L^4 + 443260L^2 + 264000) \\
& - 21280n^4L^8(127L^4 + 1726L^2 + 3900) - 31008n^2L^{12}(57L^2 + 350) - 930240L^{16}), \tag{88}
\end{aligned}$$

$$\begin{aligned}
\epsilon_9(n^2, L^2) = & \frac{n^{16}}{4194304} (408787995n^{18} + 15278638650n^{16} - 341n^{14}(416518L^4 - 289086315) \\
& - 290n^{12}(322478L^6 + 13572891L^4 - 508132290) \\
& - 9n^{10}(4892853L^8 + 244792100L^6 + 1786929620L^4 - 4297880080) \\
& - 1300n^8L^4(16555L^6 + 663164L^4 + 5343868L^2 + 8175108) \\
& - 184n^6L^6(68435L^6 + 1771020L^4 + 10489824L^2 + 12000000) \\
& - 170016n^4L^{10}(2160 + 733L^2 + 49L^4) - 769120n^2L^{14}(7L^2 + 48) - 2615008L^{18}), \tag{89}
\end{aligned}$$

$$\begin{aligned}
\epsilon_{10}(n^2, L^2) = & -\frac{n^{18}}{33554432} (9724330239n^{20} + 469170488020n^{18} \\
& - 21n^{16}(175194855L^4 - 203670581987) - 1122n^{14}(2123874L^6 + 121493988L^4 - 9238989815) \\
& - 1023n^{12}(1005969L^8 + 75201300L^6 + 829658295L^4 - 5866391024) \\
& - 116n^{10}(3765438L^{10} + 248096745L^8 + 3398870376L^6 + 9733541337L^4 - 3239200000) \\
& - 27n^8L^4(8514023L^8 + 379043252L^6 + 4451273932L^4 + 13108880480L^2 + 6339200000) \\
& - 2340n^6L^8(63925L^6 + 1738820L^4 + 13359364L^2 + 23240000) \\
& - 215280n^4L^{12}(483L^4 + 7716L^2 + 28000) - 33153120n^2L^{16}(2L^2 + 15) - 29995680L^{20}). \tag{90}
\end{aligned}$$

The dependence on λ of the ground-state energy is shown in Fig. 1 for calculations up to λ^k with $k = 3, 6, 9$. Clearly, the series for the energy levels have small convergence radius. However, the $\epsilon_{nl}(\lambda)$ function can be reconstructed from its Taylor series using the Padé approximants $[M/N](\lambda)$. The actual value of the function lies between the $[N/N](\lambda)$ and the $[(N+1)/N](\lambda)$. We can use this result to estimate the uncertainty in the reconstruction which is especially important for large values of λ . Unless stated explicitly, in the following we will use the reconstruction of the energy levels with $N = 150$, which requires a calculation of the corresponding series to order λ^{301} . This yields a confident reconstruction of the energy levels, for very large values of λ . The reconstructed energy functions up to $n = 4$ are also shown in Fig. 1. Notice that, in contrast to screening nonconfining potentials like the Yukawa and Hulthén's potentials, in the case of the Cornell potential, for a given n , the energy $\epsilon_{nl}(\lambda)$ decreases with l . This is a surprising result product of the confining linear term in the potential.

A reference point for large values of λ is the critical value λ_{cr} defined by $\epsilon(\lambda_{cr}) = 0$. Unlike the Yukawa and Hulthén potentials, this is not a ionization point because we still have bound states above this critical value for the Cornell potential. In Table I we show the results for the critical values of λ for the lowest nl levels as calculated with $N = 5$. The quoted uncertainties correspond to the difference between the $[(N+1)/N]$ and $[N/N]$ Padé approximants. These uncertainties in the reconstruction are easily reduced using higher N , obtaining very precise values for λ_{cr} , but for the Cornell potential these values are not so important thus we quote values with $N = 5$ just to have a clear idea of these reference values.

Our complete solution allows us to study also the radial probabilities for every state. Similarly to the energies, the eigenstates are convergent only for small values of λ but they can be reconstructed from the series in λ using the Padé approximants. In this case the reconstruction is more resource demanding. In Fig. 2 we plot the probabilities for the lowest-lying states reconstructed with the $[5/5]$ Padé

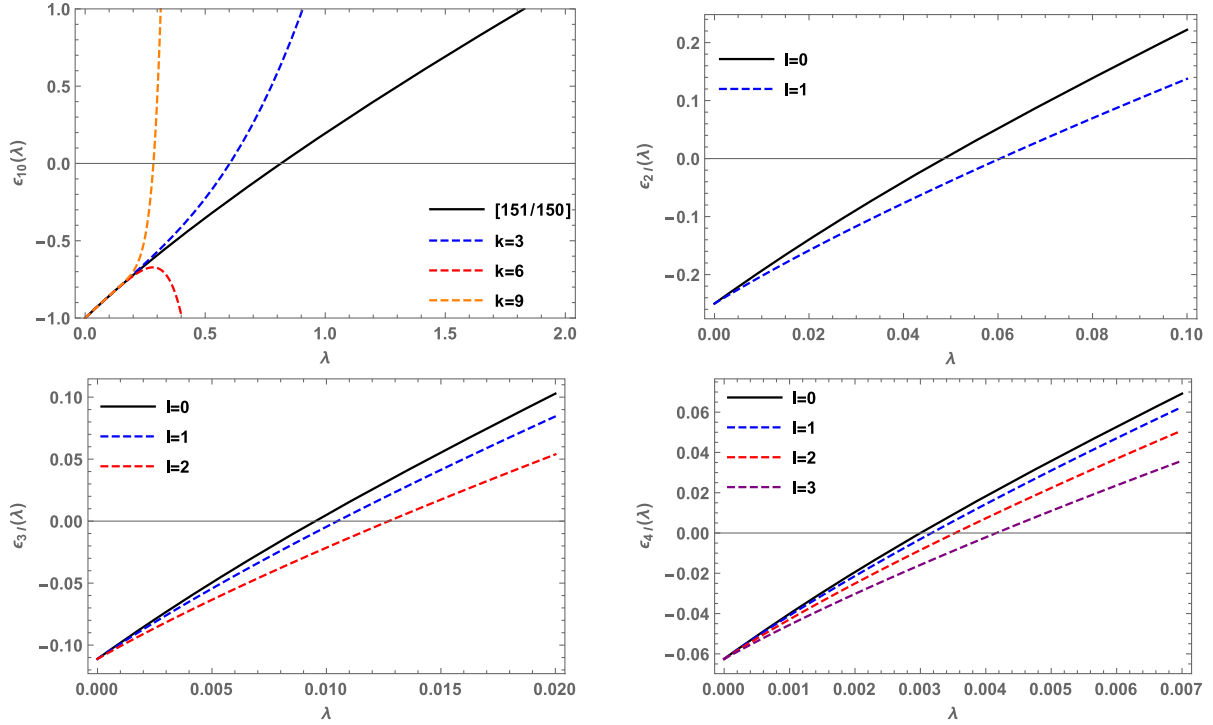


FIG. 1. Energy dependence on λ of the lowest energy levels of the Cornell potential reconstructed with the $[151/150](\lambda)$ Padé approximant. For the ground-state we also show results for the series calculated up to $\mathcal{O}(\lambda^k)$ for $k = 3, 6, 9$. Notice that ϵ_{nl} decreases with l .

approximant, for λ close to the critical value in each case, together with the result for the Coulomb-like case corresponding to $\lambda = 0$. We can see in these plots the second surprising result in the solutions; *Radial probabilities have the same form as the Coulomb-like probabilities, the main peaks grow with n , but are shifted to smaller radius by the linear term.* Instead of the conventional delocalization of states produced by nonconfining screening potential, the confining Cornell potential produces more compact states as we increase the value of λ . It is interesting that the complete solution to the Cornell potential reveals that even for values of λ as large as the critical values, the shifts are however small and the peaks remain close to the Coulomb-like values. However, in practical applications of these results it will be important to assess if the physical values of λ are around the critical values or not. We will show in the next section that for heavy quarkonium applications, except for the ground state, the physical value of λ is much larger than its critical value.

Another interesting result arising in the complete analytical solution of the nonconfining screening Yukawa and Hulthén potentials in Refs. [34–36], is the phenomena of crossing of energy levels. For the Yukawa potential this phenomena starts for $n = 4$ where, for specific values of the screening parameter close to the critical values, it happens that $\epsilon_{43} \geq \epsilon_{50}$. It is interesting to study if a similar effect exists for the Cornell potential. The energy levels for the Cornell potential are shown in Fig. 3 for $n = 1, 2, 3, 4$. We

find that the crossing phenomena for the Cornell potential starts with the $\epsilon_{43}(\lambda)$ level which for $\lambda \approx 0.4$ crosses with the $\epsilon_{30}(\lambda)$.

III. STRUCTURE OF HEAVY QUARKONIUM FROM THE CORNELL POTENTIAL

There are many remarkable qualitative and quantitative conclusions that can be drawn on the heavy quarkonium systems from the analytical solutions obtained in the previous section. When applied to heavy quarkonium, the solutions for the energy, $E_{nl}(\lambda)$, yield the analogous ones of the Bohr levels of the hydrogen atom. There are striking differences however, starting from the fact that nonperturbative strong interactions mimicked by the linear coupling breaks the l -degeneracy of the energy levels for a given n and produces the inverted spectrum described in the previous section.

In this section we address the predictions of our complete solution to the Cornell potential for the heavy quarkonium structure. In the following we will use the high-energy physics customary natural units, i.e., we will set $\hbar = c = 1$. We start with the typical distance scale associated to the strong coupling constant. For heavy quarkonium physics, the parameter α is related to the QCD coupling α_s as

$$\alpha = \frac{4}{3} \alpha_s. \quad (91)$$

TABLE I. Critical values of λ for the Cornell potential, calculated using the reconstructed functions $\epsilon_{nl}(\lambda)$.

n	l	λ_{cr}
1	0	0.8157(7)
n	l	λ_{cr}
2	0	0.048585(5)
2	1	0.06042(2)
n	l	λ_{cr}
3	0	0.0095239(5)
3	1	0.0104780(8)
3	2	0.0127192(5)
n	l	λ_{cr}
4	0	0.0030054(1)
4	1	0.0031703(1)
4	2	0.0035308(3)
4	3	0.0041608(5)
n	l	λ_{cr}
5	0	0.00122952(2)
5	1	0.00127220(3)
5	2	0.00136244(5)
5	3	0.0015114(1)
5	4	0.0017395(2)
n	l	λ_c
6	0	0.00059256(1)
6	1	0.00060674(1)
6	2	0.00063623(1)
6	3	0.00068347(2)
6	4	0.00075267(4)
6	5	0.0008506(1)
n	l	λ_c
7	0	0.000319724(4)
7	1	0.000325324(5)
7	2	0.000336847(6)
7	3	0.00035498(1)
7	4	0.00038086(1)
7	5	0.00041625(2)
7	6	0.00046379(3)
n	l	λ_c
8	0	0.000187368(2)
8	1	0.000189875(2)
8	2	0.000194997(3)
8	3	0.000202966(4)
8	4	0.000214156(5)
8	5	0.00022912(1)
8	6	0.000248667(1)
8	7	0.00027394(2)
n	l	λ_c
9	0	0.000116953(1)
9	1	0.000118186(1)

(Table continued)

TABLE I. (Continued)

n	l	λ_c
9	2	0.000120696(2))
9	3	0.000124572(2)
9	4	0.000129955(4)
9	5	0.000137047(3)
9	6	0.000146139(5)
9	7	0.00015762(1)
9	8	0.00017204(1)

The QCD Bohr radius arising from this coupling constant and denoted in the following as a_Q , is the typical length scale for quarkonium and considering Eq. (91) and the fact that for heavy quarkonium the reduced mass is $\mu = m_Q/2$ we get

$$a_Q = \frac{3}{2} \frac{1}{m_Q \alpha_s(m_Q)}. \quad (92)$$

For heavy quarkonium, the dimensionless coupling λ in Eq. (5) is related to the QCD coupling constant α_s and the string tension σ as

$$\lambda = \frac{27}{8} \frac{\sigma}{m_Q^2 \alpha_s^3}. \quad (93)$$

The masses of heavy quarkonium states can be written to leading order in a nonrelativistic expansion in terms of the eigenvalues of the Cornell potential as

$$\begin{aligned} M(n^{2S+1}L_J)(m_Q, \alpha_s(m_Q), \lambda_Q) \\ = 2m_Q + \frac{4}{9} m_Q \alpha_s^2(m_Q) \epsilon_{nl}(\lambda_Q), \end{aligned} \quad (94)$$

where m_Q , $\alpha_s(m_Q)$ and λ_Q denote the physical values of the corresponding quantities for the heavy quark Q . Notice that for $\lambda = 0$ the energy levels reduce to the Coulomb values $\epsilon_{nl}(0) = -1/n^2$. Our solution allows to go in a controlled manner from the perturbative Coulomb values to the nonperturbative region mimicked by nonzero values of the normalized string tension λ .

We remark that we have two well-defined scales in Eq. (94),

$$\mu_Q(m_Q) = 2m_Q, \quad (95)$$

$$\mu_B(m_Q) = \frac{4}{9} m_Q \alpha_s^2(m_Q). \quad (96)$$

The scale μ_Q is the scale of the perturbative effects while $\mu_B(m_Q)$ corresponds to the reference energy scale for the Bohr-like levels of heavy quarkonium.

The outstanding result of our complete analytical solution to the Cornell potential is the inverted spectrum, i.e., a spectrum where, for a given n , states with higher values of l have a lower mass. This pattern is counterintuitive in the light of results for the nonconfining screening potentials like

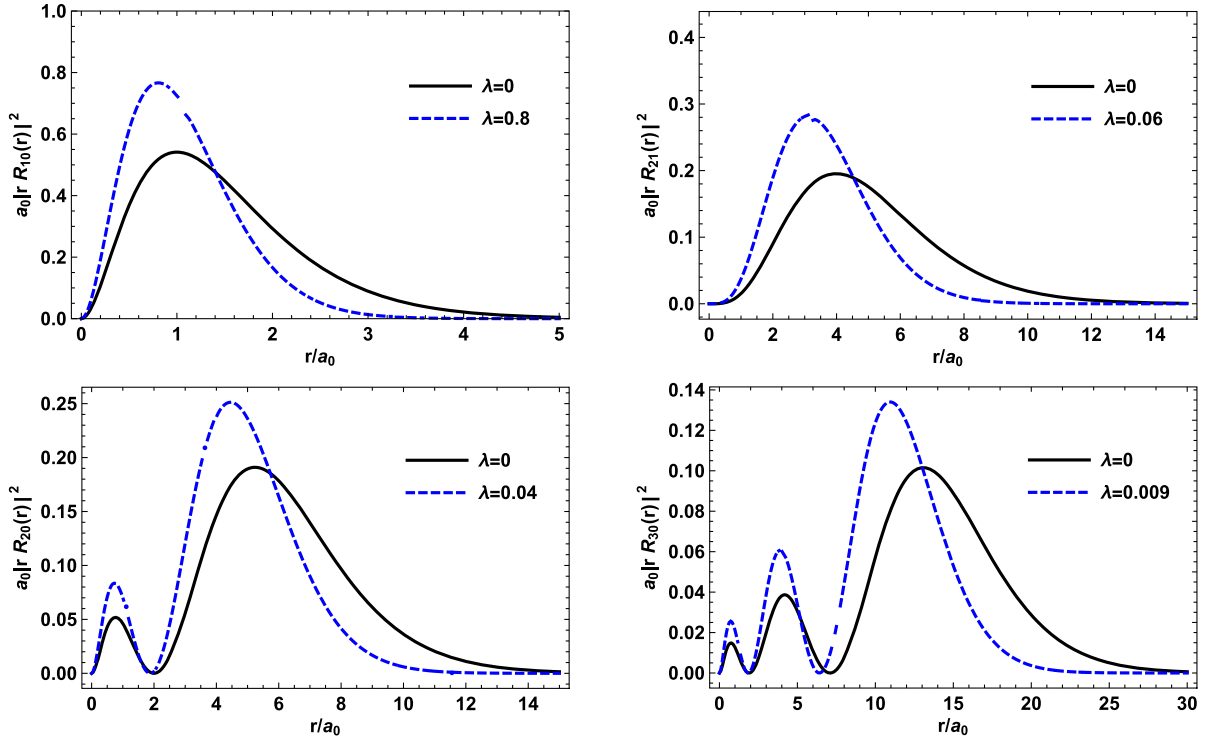


FIG. 2. Reconstruction of the probabilities for the lowest-lying states of the Cornell potential with the $[5/5](\lambda)$ Padé approximant, for values of λ close to the critical values for each state and comparison with the Coulomb-like states ($\lambda = 0$).

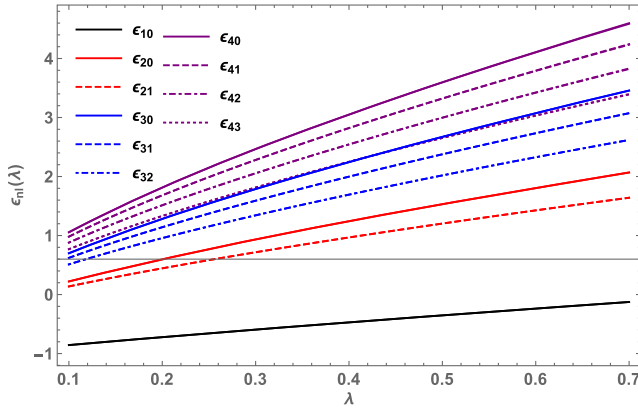


FIG. 3. Energy levels, $\epsilon_{nl}(\lambda)$, of the Cornell potential for $n = 1, 2, 3, 4$. For $\lambda \approx 0.4$ the $\epsilon_{43}(\lambda)$ level crosses with the $\epsilon_{30}(\lambda)$ level.

the Yukawa and Hulthén potentials [34–36], and its comparison against the physical heavy quarkonium spectrum is a crucial test for the Cornell potential. With this aim, we collected from the Review of Particle Properties [37], those states considered as “well-established” and ordered them in ascending order of the mass values for bottomonium states in Table II and for charmonium states in Table III.

We can see in Table II that for $n = 2$, the $l = 0$ states, $\Upsilon(2S)$ and $\eta_b(2S)$, are heavier than the $l = 1$ states, $\chi_{b2}(1P)$, $h_b(1P)$, $\chi_{b1}(1P)$, $\chi_{b0}(1P)$. Similar results hold for $n = 3$ where the $l = 0$ state, $\Upsilon(3S)$ [the $\eta_b(3S)$ state is missing], is heavier than the $l = 1$ states, $\chi_{b2}(2P)$,

$h_b(2P)$, $\chi_{b1}(2P)$, $\chi_{b0}(2P)$. In turn, these $l = 1$ states are heavier than the only $l = 2$ state so far discovered in this level, the $\Upsilon_2(1D)$ which is a 3^3D_2 state. The so far discovered bottomonium states in the $n = 4$ level follow the same pattern. *It is clear from the mass values in Table II that the physical spectrum of bottomonium follows the inverted spectrum pattern of the Bohr-like levels predicted by the complete analytical solution to the Cornell potential.*

The same pattern is clearly seen for the $n = 2$ level of charmonium. Indeed from the experimental mass values in Table III, we can see that in the $n = 2$ level, the $l = 0$ states $\psi(2S)$ and $\eta_c(2S)$ states are heavier than the $l = 1$ states $\chi_{c2}(1P)$, $h_c(1P)$, $\chi_{c1}(1P)$, $\chi_{c0}(1P)$. The identification of the $n^{2S+1}L_J$ charmonium states for $n \geq 3$ is still unclear, but we can see that a picture consistent with the inverted spectrum is obtained if we consider the $\psi(4040)$ as the 3^3S_1 state and the $\psi(3842)$, $\psi(3823)$, and $\psi(3770)$ states as the 3^3D_3 , 3^3D_1 , and 3^3D_1 charmonium states, although it has been claimed that the $\psi(3770)$ contains a sizeable tetraquark component [38,39].

An important physical quantity in heavy quarkonium physics is the size of a given quarkonium configuration. This is important information for effective theories of QCD because it is precisely the inverse radius which fixes the scale of the soft modes. For the Coulomb interaction this information is usually obtained from the mean value of the inverse radius. With our complete analytical solution we can calculate the mean value of the inverse radius

TABLE II. Experimental results for the spectrum of the bottomonium [37]. All the states in this table are considered as well-established by the Particle Data Group except for the $\eta_b(2S)$.

State	n	$^{2S+1}L_J$	M_{exp} (GeV)
$\Upsilon(4S)$	4	3S_1	10.5794(1)
$\chi_{b2}(3P)$	4	3P_2	10.5240(8)
$\chi_{b1}(3P)$	4	3P_1	10.5134(7)
$\Upsilon(3S)$	3	3S_1	10.3551(5)
$\chi_{b2}(2P)$	3	3P_2	10.26865(72)
$h_b(2P)$	3	1P_1	10.2598(12)
$\chi_{b1}(2P)$	3	3P_1	10.25546(72)
$\chi_{b0}(2P)$	3	3P_0	10.2325(9)
$\Upsilon_2(1D)$	3	3D_2	10.1637(14)
$\Upsilon(2S)$	2	3S_1	10.0234(5)
$\eta_b(2S)$	2	1S_0	9.999(4)
$\chi_{b2}(1P)$	2	3P_2	9.91221(57)
$h_b(1P)$	2	1P_1	9.8993(8)
$\chi_{b1}(1P)$	2	3P_1	9.89278(40)
$\chi_{b0}(1P)$	2	3P_0	9.85944(73)
$\Upsilon(1S)$	1	3S_1	9.46040(10)
$\eta_b(1S)$	1	1S_0	9.3987(20)

normalized to the strong Bohr radius for every n, l heavy quarkonium configuration as

$$a_Q r_{nl}^{-1}(\lambda) = \langle nl | \frac{a_Q}{r} | nl \rangle = a_Q \int_0^\infty dr r |R_{nl}(r, \lambda)|^2 \equiv f_{nl}^{(-1)}(\lambda). \quad (97)$$

We remark that the mean value of the inverse radius normalized by the strong Bohr radius is a function that depends only on the normalized string tension λ . In Fig. 4 we plot $f^{(-1)}(\lambda)$ for the lowest-lying heavy quarkonium nl states, calculated with the [101/100] Padé approximant. We can see in this plot that $r_{nl}^{-1}(\lambda)$ goes from its l -independent Coulomb value $r_{nl}^{-1}(0) = 1/n^2 a_Q$, to a l -dependent value for the physical λ_Q , which increases with increasing λ . We conclude from this plot that the effective radius of heavy quarkonium states grows with n but this growing is significantly lower than the Coulomb $n^2 a_Q$ behavior, i.e., heavy quarkonium systems are more compact than Coulomb-like systems given by the perturbative term.

A. Quantitative Bohr-like heavy quarkonium systems

The qualitative picture of quarkonium structure agrees with the measured heavy quarkonium spectrum and it would be interesting to do a more quantitative analysis of the predictions in Eq. (94). A first estimate of the physical values of the parameters and involved quantities can be obtained considering that the Borh-like levels $\epsilon_{nl}(\lambda)$ correspond to the average values of all the allowed m values in this level. We expect relativistic corrections to be important

TABLE III. Charmonium spectrum collected from the Review of Particle Properties [37]. We list only those states considered as well-established by the Particle Data Group. The $^{2S+1}L_J$ assignment in parenthesis still need to be confirmed.

State	n	$^{2S+1}L_J$	M_{exp} (GeV)
$\psi(4660)^a$	?	?	4.641(10)
$\psi(4415)$	?	3S_1	4.415(5)
$\psi(4360)^a$	?	?	4.374(7)
$\chi_{c1}(4274)^a$	?	?	$4.286^{+0.008}_{-0.009}$
$\psi(4230)$	?	3S_1	4.2221(23)
$\psi(4160)$	?	3S_1	4.191(5)
$\chi_{c1}(4140)^a$	?	?	4.1465(30)
$\psi(4040)$	?	3S_1	4.040(4)
$\chi_{c2}(3930)$	3	3P_2	3.9225(10)
$\chi_{c0}(3915)^b$	?	?	3.9221(18)
$\chi_{c1}(3872)^a$	?	?	3.87164(6)
$\psi(3842)$	?	$(^3D_3)$	3.84271(20)
$\psi(3823)$	?	$(^3D_2)$	3.82351(34)
$\psi(3770)$	?	$(^3D_1)$	3.7737(7)
$\psi(2S)$	2	3S_1	3.686097(11)
$\eta_c(2S)$	2	1S_0	3.6377(9)
$\chi_{c2}(1P)$	2	3P_2	3.55617(7)
$h_c(1P)$	2	1P_1	3.52537(14)
$\chi_{c1}(1P)$	2	3P_1	3.51067(5)
$\chi_{c0}(1P)$	2	3P_0	3.41471(30)
$J/\psi(1S)$	1	3S_1	3.096900(6)
$\eta_c(1S)$	1	1S_0	2.9841(4)

^aCandidates to be exotic states.

^bStates are in doubt in their J^P assignment.

for real heavy quarkonium systems thus the estimates in the remaining of this subsection must be considered only as a starting point toward the description of real heavy quarkonium systems.

For a given n , the average values of the masses of the $l = 0, 1$ states are given by

$$\bar{M}_{n0}(\bar{\lambda}_Q) = \frac{1}{4}(M[n^1S_0] + 3M[n^3S_1]), \quad (98)$$

$$\bar{M}_{n1}(\bar{\lambda}_Q) = \frac{1}{12}(M[n^3P_0] + 3M[n^3P_1] + 5M[n^3P_2] + 3M[n^1P_1]), \quad (99)$$

where hereafter we use a bar for the physical quantities extracted from the physical average masses.

From the values for physical quarkonium in Tables II and III, we can see that we have complete experimental data only for the $n = 1$ and $n = 2$ levels in both quarkonium sectors, thus we will use this data. We get the following average central values for the $n = 1, 2$ bottomonium and charmonium levels:

$$\bar{M}_{10}(\bar{\lambda}_b) = 9444.98 \text{ MeV}, \quad \bar{M}_{10}(\bar{\lambda}_c) = 3068.70 \text{ MeV}, \quad (100)$$

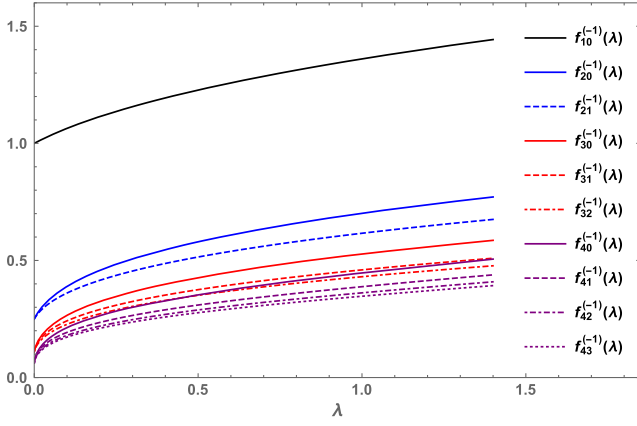


FIG. 4. Heavy quarkonium inverse mean radius in units of the Bohr radius, $\langle n | (\frac{r}{a_Q})^{-1} | n \rangle$, for the lowest-lying states.

$$\bar{M}_{20}(\bar{\lambda}_b) = 10017.30 \text{ MeV}, \quad \bar{M}_{20}(\bar{\lambda}_c) = 3674.00 \text{ MeV}, \quad (101)$$

$$\bar{M}_{21}(\bar{\lambda}_b) = 9899.73 \text{ MeV}, \quad \bar{M}_{21}(\bar{\lambda}_c) = 3525.31 \text{ MeV}. \quad (102)$$

It is convenient for the purposes of this subsection to rewrite Eq. (94) as

$$\bar{M}_{nl}(\bar{\lambda}_Q) = \mu_Q(\bar{m}_Q) + \mu_B(\bar{m}_Q) \epsilon_{nl}(\bar{\lambda}_Q). \quad (103)$$

The Bohr level values of $\bar{\lambda}_Q$, can be obtained from these relations for $n = 1, 2$ as the solution to

$$\frac{\epsilon_{20}(\bar{\lambda}_Q) - \epsilon_{10}(\bar{\lambda}_Q)}{\epsilon_{20}(\bar{\lambda}_Q) - \epsilon_{21}(\bar{\lambda}_Q)} = \frac{\bar{M}_{20}(\bar{\lambda}_Q) - \bar{M}_{10}(\bar{\lambda}_Q)}{\bar{M}_{20}(\bar{\lambda}_Q) - \bar{M}_{21}(\bar{\lambda}_Q)}. \quad (104)$$

The experimental values for the ratio on the right-hand side are obtained from the average values in Eqs. (100)–(102) as

$$\begin{aligned} \frac{\bar{M}_{20}(\bar{\lambda}_b) - \bar{M}_{10}(\bar{\lambda}_b)}{\bar{M}_{20}(\bar{\lambda}_b) - \bar{M}_{21}(\bar{\lambda}_b)} &= 4.868 \pm 0.037, \\ \frac{\bar{M}_{20}(\bar{\lambda}_c) - \bar{M}_{10}(\bar{\lambda}_c)}{\bar{M}_{20}(\bar{\lambda}_c) - \bar{M}_{21}(\bar{\lambda}_c)} &= 4.071 \pm 0.005. \end{aligned} \quad (105)$$

Solving Eq. (104) for bottomonium and charmonium, we get the following estimates of the central values of the normalized string tension, extracted from the Bohr-like (purely nonrelativistic) heavy quarkonium levels:

$$\bar{\lambda}_b = 0.884, \quad \bar{\lambda}_c = 1.694. \quad (106)$$

Although these values are natural in the sense that for dimensionless parameters we expect $\mathcal{O}(1)$ values, we must consider them as gross estimates of the physical values, until we have clear idea on the size of the relativistic

corrections. The lack of estimates of the size of these corrections prevent a serious calculation of the uncertainties in these quantities at this point, this is the reason why we refrain from quoting uncertainties in these quantities. Meanwhile, pushing forward this purely nonrelativistic quantitative picture, we can now obtain from the same data the gross values of the scales $\mu_Q(\bar{m}_Q)$ and $\mu_B(\bar{m}_Q)$, which are given by

$$\mu_B(\bar{m}_Q) = \frac{\bar{M}_{20}(\bar{\lambda}_Q) - \bar{M}_{10}(\bar{\lambda}_Q)}{\epsilon_{20}(\bar{\lambda}_Q) - \epsilon_{10}(\bar{\lambda}_Q)}, \quad (107)$$

$$\mu_Q(\bar{m}_Q) = \frac{\epsilon_{20}(\bar{\lambda}_Q) \bar{M}_{10}(\bar{\lambda}_Q) - \epsilon_{10}(\bar{\lambda}_Q) \bar{M}_{20}(\bar{\lambda}_Q)}{\epsilon_{20}(\bar{\lambda}_Q) - \epsilon_{10}(\bar{\lambda}_Q)}. \quad (108)$$

Our solutions yield the following values for the energy levels:

$$\epsilon_{10}(\bar{\lambda}_b) = 0.073, \quad \epsilon_{10}(\bar{\lambda}_c) = 0.875, \quad (109)$$

$$\epsilon_{20}(\bar{\lambda}_b) = 2.527, \quad \epsilon_{20}(\bar{\lambda}_c) = 4.299, \quad (110)$$

$$\epsilon_{21}(\bar{\lambda}_b) = 2.014, \quad \epsilon_{21}(\bar{\lambda}_c) = 3.458. \quad (111)$$

The values obtained for the energy scales at the Bohr values of the normalized string tension are

$$\mu_B(\bar{m}_b) = 233.25 \text{ MeV}, \quad \mu_B(\bar{m}_c) = 176.80 \text{ MeV}, \quad (112)$$

$$\mu_Q(\bar{m}_b) = 9427.92 \text{ MeV}, \quad \mu_Q(\bar{m}_c) = 2914.00 \text{ MeV}. \quad (113)$$

These values for the physical scales yields the following values for the strong coupling constant and heavy quark masses

$$\bar{m}_b = 4713.96 \text{ MeV}, \quad \alpha_s(\bar{m}_b) = 0.3337, \quad (114)$$

$$\bar{m}_c = 1457.0 \text{ MeV}, \quad \alpha_s(\bar{m}_c) = 0.5225. \quad (115)$$

We insist in that these values are obtained in an purely nonrelativistic calculation and should be considered only as a first approximation to the real values. We can have an idea of the variations using instead the data on the masses of the 1^3S_1 , 2^3S_1 , and 3^3S_1 , for the bottomonium states to fix $\bar{\lambda}_b$, which yields $\bar{\lambda}_b \simeq 0.4$ and similar values for $\alpha_s(\bar{m}_b)$ and $\mu_Q(\bar{m}_b), \mu_B(\bar{m}_b)$. This procedure, however, would be inconsistent because the Bohr-like levels ϵ_{n0} must correspond to the average of the n^3S_1 and n^1S_0 , but it exhibits that the extraction of the physical values of λ_Q may be sensible to corrections to the Cornell potential.

Once fixed the Bohr values of $\bar{\lambda}_Q$ and of the energy scales, our solution yields the following values for the energy of the $n = 3, 4$ excited states:

$$\epsilon_{30}(\bar{\lambda}_b) = 4.121, \quad \epsilon_{30}(\bar{\lambda}_c) = 6.666, \quad (116)$$

$$\epsilon_{31}(\bar{\lambda}_b) = 3.663, \quad \epsilon_{31}(\bar{\lambda}_c) = 5.919, \quad (117)$$

$$\epsilon_{32}(\bar{\lambda}_b) = 3.128, \quad \epsilon_{32}(\bar{\lambda}_c) = 5.076, \quad (118)$$

$$\epsilon_{40}(\bar{\lambda}_b) = 5.448, \quad \epsilon_{40}(\bar{\lambda}_c) = 8.762, \quad (119)$$

$$\epsilon_{41}(\bar{\lambda}_b) = 5.024, \quad \epsilon_{41}(\bar{\lambda}_c) = 8.067, \quad (120)$$

$$\epsilon_{42}(\bar{\lambda}_b) = 4.537, \quad \epsilon_{42}(\bar{\lambda}_c) = 7.292, \quad (121)$$

$$\epsilon_{43}(\bar{\lambda}_b) = 4.028, \quad \epsilon_{43}(\bar{\lambda}_c) = 6.488, \quad (122)$$

and from Eq. (103) the following predictions of the Cornell potential for the average masses of the $n = 3, 4$ excited states are obtained,

$$\bar{M}_{30}(\bar{\lambda}_b) = 10389.10 \text{ MeV}, \quad \bar{M}_{30}(\bar{\lambda}_c) = 4092.46 \text{ MeV}, \quad (123)$$

$$\bar{M}_{31}(\bar{\lambda}_b) = 10282.20 \text{ MeV}, \quad \bar{M}_{31}(\bar{\lambda}_c) = 3960.52 \text{ MeV}, \quad (124)$$

$$\bar{M}_{32}(\bar{\lambda}_b) = 10157.40 \text{ MeV}, \quad \bar{M}_{32}(\bar{\lambda}_c) = 3811.48 \text{ MeV}, \quad (125)$$

$$\bar{M}_{40}(\bar{\lambda}_b) = 10698.70 \text{ MeV}, \quad \bar{M}_{40}(\bar{\lambda}_c) = 4463.19 \text{ MeV}, \quad (126)$$

$$\bar{M}_{41}(\bar{\lambda}_b) = 19599.80 \text{ MeV}, \quad \bar{M}_{41}(\bar{\lambda}_c) = 4340.29 \text{ MeV}, \quad (127)$$

$$\bar{M}_{42}(\bar{\lambda}_b) = 10486.20 \text{ MeV}, \quad \bar{M}_{42}(\bar{\lambda}_c) = 4203.22 \text{ MeV}, \quad (128)$$

$$\bar{M}_{43}(\bar{\lambda}_b) = 10367.40 \text{ MeV}, \quad \bar{M}_{43}(\bar{\lambda}_c) = 4061.04 \text{ MeV}. \quad (129)$$

From data in Table II, the only available average value beyond those in Eqs. (100)–(102) used to fix the free parameters, is the average mass of the $n = 3, l = 1$ level, where we have data on the masses of all the members of the corresponding multiplet. Other multiplets have missing states. From data in Table II we obtain $\bar{M}_{31}(\bar{\lambda}_b)_{\text{exp}} = 10260.10 \text{ MeV}$. This central value is 0.2% off the central value prediction in Eq. (124). This corresponds to a difference of the order of 22 MeV which should be the order of magnitude of the corrections to the Cornell potential, which include relativistic corrections. In this concern, we can estimate from our solutions the average v^2 for the different heavy quarkonium configurations, which can give us an idea of the size of relativistic

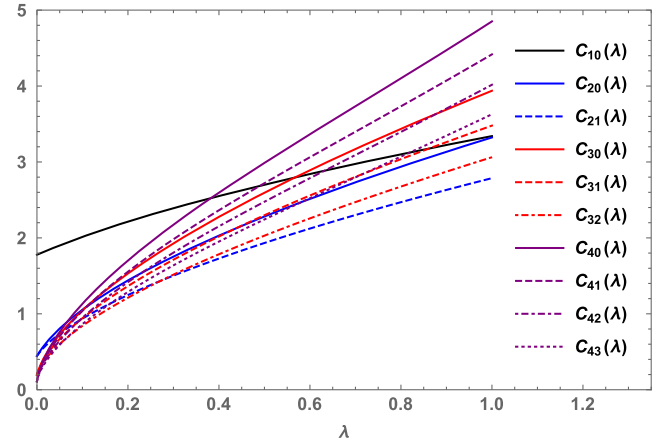


FIG. 5. Proportionality constant $C_{nl}(\lambda)$ between $\langle nl|v^2|nl\rangle$ and α_s^2 as a function of λ for the lowest-lying states.

corrections. This observable is given by

$$\langle nl|v^2|nl\rangle = \frac{4}{m_Q} \langle nl|H_0 + \frac{4\alpha_s}{3r} - \sigma r|nl\rangle = \alpha_s^2 C_{nl}(\lambda), \quad (130)$$

with

$$C_{nl}(\lambda) = \frac{16}{9} (\epsilon_{nl}(\lambda) + 2f_{nl}^{(-1)}(\lambda) - \lambda f_{nl}^{(1)}(\lambda)), \quad (131)$$

where

$$f_{nl}^{(k)}(\lambda) \equiv \langle nl|\left(\frac{r}{a_Q}\right)^k|nl\rangle = a_Q^{-k} \int_0^\infty dr r^{2+k} |R_{nl}(r)|^2. \quad (132)$$

are dimensionless functions depending only on λ .

Using the virial theorem it can be shown that for the Cornell potential,

$$\epsilon_{nl}(\lambda) = -f_{nl}^{(-1)}(\lambda) + \frac{3}{2} \lambda f_{nl}^{(1)}(\lambda), \quad (133)$$

and the proportionality constant can also be written as

$$C_{nl}(\lambda) = \frac{8}{9} (2f_{nl}^{(-1)}(\lambda) + \lambda f_{nl}^{(1)}(\lambda)). \quad (134)$$

Equations (130) and (134) show that the average value of v^2 is indeed proportional of α_s^2 as expected from the NRQCD counting rules. However, the proportionality constant depends in a complicated manner on the string tension. The detail of this dependence can be obtained from our complete solution to the Cornell potential and it is shown in Fig. 5 where we plot the proportionality constant C_{nl} as a function of λ for the lowest-lying $n = 1, 2, 3, 4$ states, calculated with the [101/100] Padé approximant. We can see in this plot that the proportionality constant changes its behavior from its Coulombian value $C_{nl}(0) = \frac{16}{9n^2}$ decreasing with n and l -independent, to a value around 2

which is l -dependent, for $\lambda \approx 0.4$. The general lesson from this plot is that relativistic corrections will be more important than expected for both ground and excited heavy quarkonium states, although a precise quantification of this effect must await for a more precise extraction of the physical value λ_Q from data. We devote the next section to the calculation of the leading relativistic spin-dependent corrections which in addition to modify the predictions for the Bohr levels break the m -degeneracy in $\epsilon_{nl}(\lambda)$ through the spin-dependent interactions. In preparation, we notice that there is an obvious mass hierarchy in the experimental fine splittings of heavy quarkonium which can be seen in Tables II and III. Indeed, from the measured values of the masses of the $n = 1$ and $n = 2$ levels of heavy quarkonium, systematically for the s -waves, the n^3S_1 state is heavier than the n^1S_0 state. Also, for the $n = 2$ and $n = 3$ (in the bottomonium case) p -wave states, the masses are such that systematically $M_{\chi_{b2}(nP)} > M_{h_b(nP)} > M_{\chi_{b1}(nP)} > M_{\chi_{b0}(nP)}$.

IV. TOWARDS REAL HEAVY QUARKONIUM: SPIN-DEPENDENT RELATIVISTIC CORRECTIONS

Relativistic corrections are expected from Eq. (130) to be of order α_s^4 and to yield a closer approach to real heavy quarkonium systems. The derivation of the leading relativistic corrections has been done using diverse methods [5–11,18,40] and the corresponding operators can be classified into spin-independent and spin-dependent interactions. The leading spin-independent corrections include conventional corrections to the kinetic energy and velocity-dependent terms [10,11], which shifts the Bohr-like energy levels $\epsilon_{nl}(\lambda)$ but do not remove the m -degeneracy of the l orbitals for a fixed n . It is not the aim of this work to do a complete phenomenological analysis of the relativistic corrections and we will focus only on the fine splittings in heavy quarkonium produced by the spin-dependent interactions. However, there are interesting conclusions arising from this analysis concerning the $1/m_Q$ expansion and we will consider explicitly the corrections to the kinetic energy in the spin-independent sector to illustrate the point. These corrections are induced by

$$H_K = -\frac{p^4}{4m_Q^3} = -\frac{1}{4m_Q} \left(H_0 + \frac{4\alpha_s}{3r} - \frac{8}{27}m_Q^2\alpha_s^3\lambda r \right)^2. \quad (135)$$

The corrections to E_{nl} due to this term, as calculated in perturbation theory are given by

$$\begin{aligned} \Delta E_{nl}^{si} = & -\frac{4}{81}m_Q\alpha_s^4(\epsilon_{nl}^2(\lambda) + 4f_{nl}^{(-2)}(\lambda) + \lambda^2 f_{nl}^{(2)}(\lambda) \\ & + 4\epsilon_{nl}(\lambda)f_{nl}^{(-1)}(\lambda) - 2\lambda\epsilon_{nl}(\lambda)f_{nl}^{(1)}(\lambda) - 4\lambda). \end{aligned} \quad (136)$$

A. Spin-dependent interactions

The proper description of the physical spectrum of heavy quarkonium requires to consider the spin-dependent interactions responsible for the splitting of the Bohr energy levels $\epsilon_{nl}(\lambda)$. These interactions arise from the leading relativistic corrections and for the Cornell potential are given by

$$H_{SD} = \frac{8\alpha_s}{9m_Q^2} \mathbf{S}_1 \cdot \mathbf{S}_2 \frac{\delta(r)}{r^2} + \frac{2\alpha_s \mathbf{L} \cdot \mathbf{S}}{m_Q^2 r^3} + \frac{\alpha_s}{3m_Q^2} \frac{S_{12}}{r^3} - \frac{\sigma}{2m_Q^2} \frac{\mathbf{L} \cdot \mathbf{S}}{r}, \quad (137)$$

where

$$S_{12} = 4[3(\mathbf{S}_1 \cdot \hat{\mathbf{r}})(\mathbf{S}_2 \cdot \hat{\mathbf{r}}) - \mathbf{S}_1 \cdot \mathbf{S}_2]. \quad (138)$$

The leading relativistic corrections have a $1/m_Q^2$ suppression and are expected to be small. The first term in Eq. (137) is the analogous of the hyperfine splitting in atomic physics where it is suppressed due to the small electron to proton mass ratio. Here, this term is of the same order as the remaining fine structure terms and we will denote all terms in Eq. (137) as fine-structure interactions. We can treat these interactions perturbatively due to their formal $1/m_Q^2$ suppression. To leading order we need to calculate the spin-dependent matrix elements,

$$\langle \bar{Q}Q[n^{2S+1}L_J] | O_i | \bar{Q}Q[n^{2S+1}L_J] \rangle, \quad (139)$$

for $O_1 = \mathbf{S}_1 \cdot \mathbf{S}_2 \frac{\delta(r)}{r^2}$, $O_2 = \frac{\mathbf{L} \cdot \mathbf{S}}{r^3}$, $O_3 = \frac{S_{12}}{r^3}$, and $O_4 = \frac{\mathbf{L} \cdot \mathbf{S}}{r}$. The angular momentum part yields the well-defined n -independent factors,

$$\langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle = \frac{1}{2} \left[s(s+1) - \frac{3}{2} \right], \quad (140)$$

$$\langle \mathbf{L} \cdot \mathbf{S} \rangle = \frac{1}{2} [j(j+1) - l(l+1) - s(s+1)], \quad (141)$$

$$\langle S_{12} \rangle = \frac{2[2l(l+1)s(s+1) - 3\langle \mathbf{L} \cdot \mathbf{S} \rangle - 6\langle \mathbf{L} \cdot \mathbf{S} \rangle^2]}{(2l-1)(2l+3)}. \quad (142)$$

The explicit spin factors for the lowest-lying angular momentum configurations are given in Table IV.

The corrections to E_{nl} due to the spin-dependent term to the mass of heavy quarkonium, as calculated in perturbation theory, are given by

$$\begin{aligned} \Delta E_{nl}^{sd}(\lambda) = & \frac{8\alpha_s}{9m_Q^2} \langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle \left\langle \frac{\delta(r)}{r^2} \right\rangle_{nl} \\ & + \left(\frac{2\alpha_s}{m_Q^2} \left\langle \frac{1}{r^3} \right\rangle_{nl} - \frac{\sigma}{2m_Q^2} \left\langle \frac{1}{r} \right\rangle_{nl} \right) \langle \mathbf{L} \cdot \mathbf{S} \rangle \\ & + \frac{\alpha_s}{3m_Q^2} \langle S_{12} \rangle \left\langle \frac{1}{r^3} \right\rangle_{nl}. \end{aligned} \quad (143)$$

This result can be written as

$$\Delta E_{nl}^{sd}(\lambda) = \frac{16}{243} m_Q \alpha_s^4 \left[f_{nl}^{ss}(\lambda) 4 \langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle + \left(9 f_{nl}^{(-3)}(\lambda) - \frac{3}{2} \lambda f_{nl}^{(-1)}(\lambda) \right) \langle \mathbf{L} \cdot \mathbf{S} \rangle + \frac{3}{2} f_{nl}^{(-3)}(\lambda) \langle S_{12} \rangle \right], \quad (144)$$

where

$$f_{nl}^{ss}(\lambda) = a_Q^3 \int_0^\infty dr \delta(r) |R_{nl}(r, \lambda)|^2 = a_Q^3 |R_{nl}(0, \lambda)|^2 \quad (145)$$

and $f^{(k)}(\lambda)$ are given in Eq. (132).

B. Heavy quarkonium spectrum to order α_s^4

Considering both spin-independent and spin-dependent interactions, the mass of heavy quarkonium in a $n^{2S+1}L_J$ configuration to order α_s^4 is given by

$$\begin{aligned} M[n^{2S+1}L_J] &= 2m_Q + \frac{4}{9} m_Q \alpha_s^2 \epsilon_{nl}(\lambda) - \frac{64}{243} m_Q \alpha_s^4 \Delta_{nl}^{SI}(\lambda) \\ &\quad + \frac{64}{243} m_Q \alpha_s^4 \left[f_{nl}^{ss}(\lambda) \langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle + \left(\frac{9}{4} f_{nl}^{(-3)}(\lambda) - \frac{3}{8} \lambda f_{nl}^{(-1)}(\lambda) \right) \langle \mathbf{L} \cdot \mathbf{S} \rangle + \frac{3}{8} f_{nl}^{(-3)}(\lambda) \langle S_{12} \rangle \right], \end{aligned} \quad (146)$$

with

$$\begin{aligned} \Delta_{nl}^{SI}(\lambda) &= \frac{3}{16} \epsilon_{nl}^2(\lambda) + \frac{3}{4} f_{nl}^{(-2)}(\lambda) + \frac{3}{16} \lambda^2 f_{nl}^{(2)}(\lambda) \\ &\quad + \frac{3}{8} \epsilon_{nl}(\lambda) (2 f_{nl}^{(-1)}(\lambda) - \lambda f_{nl}^{(1)}(\lambda)) - \frac{3}{4} \lambda. \end{aligned} \quad (147)$$

This is the second main result of this paper and there are several worthy remarks on Eq. (146). Firstly, notice that the formal $1/m_Q^2$ suppression of the spin-dependent interactions in Eq. (137) is canceled by the m_Q dependence of the involved matrix elements. This result arises from our choice of the Bohr radius as the reference dimension scale, thus the average value e.g., of $\langle 1/r^3 \rangle_{nl}$ must be proportional to $1/a_Q^3 \approx m_Q^3 \alpha_s^3$. This is also valid for the velocity-dependent terms in the spin-independent sector not shown here.

Secondly, with this result, we get actually a consistent expansion in powers of α_s^2 for the masses of heavy quarkonium. At leading order we simply get the perturbative physics scale $\mu_Q = 2m_Q$. At order α_s^2 , the coefficient is given by $\frac{4}{9} m_Q \epsilon_{nl}(\lambda)$ and involves nonperturbative QCD effects mimicked by the string tension λ . This coefficient is actually

different for the different quarkonium configurations. The different coefficients are calculated exactly here and involve the analytic solution for the normalized energies of the Cornell potential, $\epsilon_{nl}(\lambda)$. For the quarkonium spectrum, the nonperturbative effects manifest at order α_s^2 in the λ dependence of $\epsilon_{nl}(\lambda)$, which is responsible for the breaking of the l -degeneracy of the n -levels of the Coulomb-like interaction. The order α_s^4 corrections include spin-independent and spin-dependent interactions and their nonperturbative effects are calculated using Rayleigh-Schrödinger perturbation theory. Our approximate calculation of these effects shows that the size of nonperturbative effects at this order are modulated by the mean values of the space part of the corresponding operators which also involve the probabilities of the Cornell potential, $|R_{nl}(r, \lambda)|^2$.

These results show that, for the heavy quarkonium spectrum, the suppression of higher order terms in the non-relativistic expansion (including nonperturbative effects) depends only on the expansion in $\alpha_s^2(m_Q)$ and the whole expansion makes sense as long as $\alpha_s(m_Q)$ is still small, which we expect to be satisfied for heavy quarkonium. Notice that Eq. (146) is consistent with the power counting rules of NRQCD, where we have a perturbative scale m_Q and nonperturbative effects are suppressed by powers of $v^2 \approx \alpha_s^2(m_Q)$. In this concern, we remark that in Eq. (146) a new scale appears, given by

$$\mu_f(m_Q) = \frac{64}{243} m_Q \alpha_s^4, \quad (148)$$

which is the natural scale of the fine splittings of the heavy quarkonium spectrum.

The mass hierarchies in the fine splittings of the measured heavy quarkonium spectrum noticed at the end of the previous section can be understood from Eq. (146) and the values of average spin-dependent operators in Table IV. Indeed, notice that Eq. (146) can be written as

$$M[n^{2S+1}L_J] = M_{nl}^{SI}(\lambda) + M^{SD}[n^{2S+1}L_J](\lambda), \quad (149)$$

where the spin-independent part, $M_{nl}^{SI}(\lambda)$, contain terms of order α_s^2 and α_s^4 which, for a given n , still preserve the m -degeneracy of the l -levels. This degeneracy is broken by the spin-dependent contribution $M^{SD}[n^{2S+1}L_J](\lambda)$ which only contains terms of order α_s^4 producing the fine splittings of heavy quarkonium.

TABLE IV. Spin factors for the lowest lying angular momentum configurations.

Operator	1S_0	3S_1	1P_1	3P_0	3P_1	3P_2	1D_2	3D_1	3D_2	3D_3
$\langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle$	$-\frac{3}{4}$	$\frac{1}{4}$	$-\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$-\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\langle \mathbf{L} \cdot \mathbf{S} \rangle$	0	0	0	-2	-1	1	0	-3	-1	2
$\langle S_{12} \rangle$	0	0	0	-4	2	$-\frac{2}{5}$	0	-2	2	$-\frac{4}{7}$

From the mean values of the spin operators in Table IV, we can see that the spin-dependent contribution to the masses of the $l = 0$ states in a given level n , get contributions of the spin-spin interactions only and we obtain

$$M[n^3S_1] - M[n^1S_0] = \mu_f f_{nl}^{ss}(\lambda) > 0, \quad (150)$$

thus the n^3S_1 states are heavier than the n^1S_0 states. This pattern is exhibited by the $n = 1, 2$ levels of heavy quarkonium as discussed at the end of the previous section and we predict this behavior to be valid for every level n in both heavy quarkonium sectors.

The ordering of the 3P_J states in a given level n can also be obtained from Eq. (146). Indeed, a straightforward calculation using the spin factors Table IV yields,

$$\begin{aligned} M[n^3P_2] - M[n^3P_1] &= \mu_f \left(\frac{18}{5} f_{n1}^{(-3)}(\lambda) - \frac{3}{4} \lambda f_{n1}^{(-1)}(\lambda) \right) \\ &\equiv \mu_f F_{21}^P(n, \lambda), \end{aligned} \quad (151)$$

$$\begin{aligned} M[n^3P_1] - M[n^3P_0] &= \mu_f \left(\frac{9}{2} f_{n1}^{(-3)}(\lambda) - \frac{3}{8} \lambda f_{n1}^{(-1)}(\lambda) \right) \\ &\equiv \mu_f F_{10}^P(n, \lambda). \end{aligned} \quad (152)$$

As for the D -waves in the n th level, the mass splittings are given by

$$\begin{aligned} M[n^3D_3] - M[n^1D_2] &= \mu_f \left(\frac{81}{14} f_{n2}^{(-3)}(\lambda) - \frac{9}{8} \lambda f_{n2}^{(-1)}(\lambda) \right) \\ &\equiv \mu_f F_{32}^D(n, \lambda), \end{aligned} \quad (153)$$

$$\begin{aligned} M[n^3D_2] - M[n^3D_1] &= \mu_f \left(6 f_{n2}^{(-3)}(\lambda) - \frac{3}{4} \lambda f_{n2}^{(-1)}(\lambda) \right) \\ &\equiv \mu_f F_{21}^D(n, \lambda). \end{aligned} \quad (154)$$

In Fig. 6 we plot the functions $F_{21}^P(n, \lambda)$, $F_{10}^P(n, \lambda)$, $F_{32}^D(n, \lambda)$, $F_{21}^D(n, \lambda)$ for $n = 2, 3$, calculated with the [101/100] Padé approximant, where we can see that these functions are positive. This results yields the ordering

$$M[n^3P_2] > M[n^3P_1] > M[n^3P_0], \quad (155)$$

for $n = 2, 3$ states and

$$M[n^3D_3] > M[n^3D_2] > M[n^3D_1], \quad (156)$$

for $n = 3$. This ordering is clearly seen in the physical spectrum of bottomonium for the 2^3P_J , 3^3P_J , and 3^3D_J states. In the case of charmonium, the measured 2^3P_J states also exhibit this ordering. For $n = 3$, it requires the assignment of the 3^3D_J states to the $\psi(3842)$, $\psi(3823)$, and $\psi(3770)$ states. Notice that at the qualitative level discussed in this section, both the inverted spectrum and the

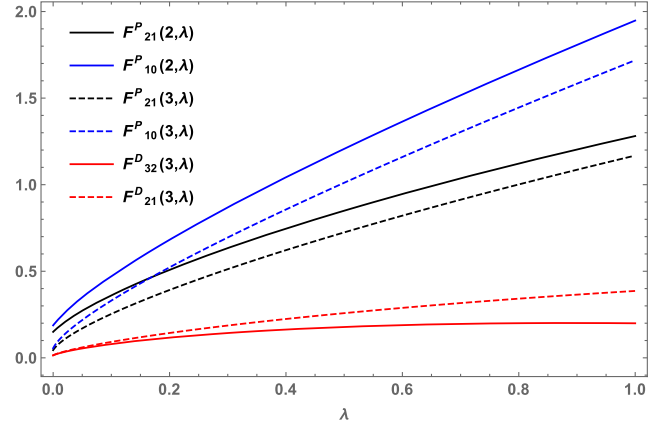


FIG. 6. Combinations $F_{21}^P(2, \lambda)$, $F_{10}^P(2, \lambda)$, $F_{21}^P(3, \lambda)$, $F_{10}^P(3, \lambda)$, $F_{32}^D(2, \lambda)$, $F_{21}^D(3, \lambda)$ appearing in Eqs. (151)–(154) for $n = 2, 3$, as functions of λ .

order α_s^4 corrections obtained here yields a consistent picture for the heavy quarkonium spectrum.

V. ASSESSING NONPERTURBATIVE EFFECTS AND PREDICTIONS FOR THE MISSING HEAVY QUARKONIUM STATES

It is not the aim of the present work to give a detailed numerical analysis of the predictions of the complete analytical solution of the Cornell potential and the leading relativistic corrections obtained here for the quarkonium spectrum, but it would be interesting to have a more precise estimate of the physical normalized string tension λ_Q than that given by the purely nonrelativistic physics in Eq. (106). In this section we will obtain it confidently from the fine splittings. It will allow us to make well-defined predictions on the mass values of some missing states and on the range of mass values where other missing heavy quarkonium states must lie, which is relevant for the experimental searches of these states.

In this concern, we can see from Eq. (150) that the fine splittings in the S -wave sectors are dictated by the fine energy scale μ_f and the value of $f_{nl}^{ss}(\lambda)$ at the physical value λ_Q . Indeed, a straightforward calculation yields,

$$M[2^3S_1] - M[2^1S_0] = \mu_f(m_Q) f_{20}^{ss}(\lambda_Q), \quad (157)$$

$$M[1^3S_1] - M[1^1S_0] = \mu_f(m_Q) f_{10}^{ss}(\lambda_Q). \quad (158)$$

Notice that these splittings are not affected by spin-independent relativistic corrections which cancel in the difference. The fine scale $\mu_f(m_Q)$ cancels in the ratio and a robust extraction of the physical value of λ_Q can be done solving,

$$\frac{M[2^3S_1] - M[2^1S_0]}{M[1^3S_1] - M[1^1S_0]} = \frac{f_{20}^{ss}(\lambda_Q)}{f_{10}^{ss}(\lambda_Q)}. \quad (159)$$

It is important to remark that the normalized wave function at the origin, $f_{n0}^{ss}(m_Q)$, actually receive considerable radiative corrections as noticed in [6]. These corrections largely cancel in the ratio of the normalized wave functions in Eq. (159) and a reliable estimate of λ_Q for heavy quarkonium can be obtained from this equation, which is independent of the α_s and m_Q factors. The ratio on the right-hand side can be obtained from the complete analytical solution to the Cornell potential and depends only on λ . The experimental value of the left-hand side of this equation can be extracted from Tables II and III and we get

$$\frac{M[\Upsilon(2S)] - M[\eta_b(2S)]}{M[\Upsilon(1S)] - M[\eta_b(1S)]} = 0.395 \pm 0.017, \quad (160)$$

$$\frac{M[\psi(2S)] - M[\eta_c(2S)]}{M[J/\psi(1S)] - M[\eta_c(1S)]} = 0.429 \pm 0.008. \quad (161)$$

The dimensionless functions $f_{nl}^{ss}(\lambda)$ are related to $R_{nl}^2(0, \lambda)$ by Eq. (145). We reconstructed $R_{10}^2(0, \lambda)$ and $R_{20}^2(0, \lambda)$ using the [151/150] and [150/150] Padé approximants. With these reconstructed functions, solving Eq. (159) we get the following values for the physical normalized string tension at the bottomonium and charmonium mass scales,

$$\lambda_b = 0.361 \pm 0.054, \quad \lambda_c = 0.482 \pm 0.033. \quad (162)$$

Notice that these values are considerably lower than those extracted from the data on the average masses of the $n = 1$, $l = 0$, and $n = 2$, $l = 0, 1$ in a purely nonrelativistic model in Eq. (106). The physical values in Eq. (162) incorporates the leading relativistic effects in a controlled α_s^2 expansion. The uncertainties in the $\mathcal{O}(\alpha_s^4)$ result in Eq. (162) do not include theoretical uncertainties due to other $\mathcal{O}(\alpha_s^6)$ terms or due to additional QCD effects, which must be eventually considered, but we expect these effects to be largely canceled in the ratios in Eqs. (160) and (161).

Once fixed the physical values of λ_Q , from our complete analytical solution we can calculate the numerical factors $\epsilon_{nl}(\lambda_Q)$ and $f_{nl}^{ss}(\lambda_Q)$ which are collected in Table V for values up to $n = 4$. To this end we use the $\epsilon_{nl}(\lambda)$ and $f_{nl}^{ss}(\lambda)$ functions reconstructed with the [151/150] and [150/150] Padé approximants. The actual value of the corresponding functions lie between these approximants and we estimate the uncertainty in the reconstruction as the ratio of the difference and the sum of these functions. We list the reconstruction uncertainties in the calculation of the physical quantities in the last column of Table V.

A. Bottomonium

The fine structure scale for bottomonium can be obtained from Eq. (150) which yields,

TABLE V. Numerical values of observables relevant for the calculation of the spectrum of the S -wave states of heavy quarkonium, obtained from the eigenvalues and eigenstates of the Cornell potential, reconstructed with the [151/150] Padé approximant from the corresponding series calculated to order λ^{301} .

Observable	Value	Rec. uncertainty
λ_b	0.361 ± 0.054	
λ_c	0.482 ± 0.033	
$\epsilon_{10}(\lambda_b)$	-0.518 ± 0.06	1.0×10^{-16}
$\epsilon_{20}(\lambda_b)$	1.124 ± 0.165	2.1×10^{-13}
$\epsilon_{30}(\lambda_b)$	2.070 ± 0.245	2.1×10^{-7}
$\epsilon_{40}(\lambda_b)$	2.828 ± 0.314	6.8×10^{-5}
$\epsilon_{10}(\lambda_c)$	-0.374 ± 0.039	7.4×10^{-17}
$\epsilon_{20}(\lambda_c)$	1.480 ± 0.094	1.1×10^{-11}
$\epsilon_{30}(\lambda_c)$	2.596 ± 0.138	1.8×10^{-6}
$\epsilon_{40}(\lambda_c)$	3.500 ± 0.176	2.7×10^{-4}
$f_{10}^{ss}(\lambda_b)$	5.681 ± 0.212	7.8×10^{-17}
$f_{20}^{ss}(\lambda_b)$	2.246 ± 0.182	2.9×10^{-13}
$f_{30}^{ss}(\lambda_b)$	1.681 ± 0.155	2.8×10^{-7}
$f_{40}^{ss}(\lambda_b)$	1.440 ± 0.141	8.0×10^{-5}
$f_{10}^{ss}(\lambda_c)$	6.140 ± 0.121	$< 1 \times 10^{-17}$
$f_{20}^{ss}(\lambda_c)$	2.634 ± 0.101	1.5×10^{-11}
$f_{30}^{ss}(\lambda_c)$	2.014 ± 0.087	2.2×10^{-6}
$f_{40}^{ss}(\lambda_c)$	1.743 ± 0.080	3.0×10^{-4}

$$\mu_f(m_b) = \frac{M[\Upsilon(1S)] - M[\eta_b(1S)]}{f_{10}^{ss}(\lambda_b)} = 10.86 \pm 0.54 \text{ MeV}. \quad (163)$$

As a cross-check, we obtain the same value if we use instead data on the $n = 2$, S -wave states.

We can see in Table II that states in the $n = 1, 2$ levels are complete and follow the inverted spectrum pattern predicted by the Cornell potential and the P -waves ordering obtained from the leading relativistic corrections. For $n \geq 3$ the states $\eta_b(nS)$ are missing. The value of the masses of these states can be obtained from Eq. (150) as

$$M[\eta_b(nS)] = M[\Upsilon(nS)] - \mu_f(m_b) f_{n0}^{ss}(\lambda_b). \quad (164)$$

Using the values for $f_{30}^{ss}(\lambda_b)$ and $f_{40}^{ss}(\lambda_b)$ quoted in Table V, and the experimental values for the masses of the $\Upsilon(3S)$ and $\Upsilon(4S)$ in Table II we predict the masses of the missing $\eta_b(3S)$ and $\eta_b(4S)$ as

$$M[\eta_b(3S)] = 10337 \pm 2 \text{ MeV}, \quad (165)$$

$$M[\eta_b(4S)] = 10564 \pm 2 \text{ MeV}. \quad (166)$$

As to the D -wave states in the $n = 3$ level with configurations 3^3D_3 , 3^1D_2 , 3^3D_2 , and 3^1D_1 , only the 3^3D_2 [the $\Upsilon_2(1D)$] has been discovered with a mass $M[\Upsilon_2(1D)] = 10163.7 \text{ MeV}$. The ordering of D -wave

states in Eq. (156) and the inverted spectrum requires the masses of the 3^3D_3 , 3^1D_2 states to be in range [10163, 10232] MeV. Similarly, the mass of the 3^1D_1 state is predicted to be in the [10023, 10163] MeV range.

Concerning the states in the $n = 4$ level of bottomonium, we have the following missing P , D , and F -wave states: $4^1P_1, 4^3P_0, 4^3D_3, 4^1D_2, 4^3D_2, 4^3D_1, 4^3F_4, 4^1F_3, 4^3F_3, 4^3F_2$. Our complete analytical solution predicts all these states to be below the $\chi_{b2}(3P)$ state, and above the $\Upsilon(3S)$ state, i.e., in the energy range [10355, 10524] MeV, with P states being heavier than the D states which in turn are more massive than the F states. In particular, the F states must be around the mass of the $\Upsilon(3S)$ state because it is precisely with these levels where the crossing phenomena starts.

B. Charmonium

The values of the masses for charmonium states listed in Table III shows that states in the $n = 1, 2$ levels are complete and follow the inverted spectrum pattern predicted by the Cornell potential. Also, the ordering of the P -waves follows the pattern predicted by the leading relativistic corrections. We can extract the value of the fine structure scale for charmonium from Eq. (150) and data on the $n = 1$, S -wave states which yields

$$\mu_f(m_c) = \frac{M[J/\psi(1S)] - M[\eta_c(1S)]}{f_{10}^{ss}(\lambda_c)} = 18.37 \pm 0.36 \text{ MeV}. \quad (167)$$

We cross-checked this result using instead data of the $n = 2$, S -wave charmonium states obtaining the same value.

In the $n = 3$ level of charmonium the first challenge is the identification of the 3^3S_1 charmonium state [the $\psi(3S)$]. The calculation of the mass of this state requires the complete numerical analysis of Eq. (146) which is beyond the scope of this paper. However, modulo corrections of order α_s^4 , the Bohr scale can be obtained from two successive Bohr levels, e.g., from the $2S$ and $1S$ bottomonium states we get

$$\mu_B(m_b) \approx \frac{M[\Upsilon(2S)] - M[\Upsilon(1S)]}{\epsilon_{20}(\lambda_b) - \epsilon_{10}(\lambda_b)} = 343 \text{ MeV}. \quad (168)$$

Similar values are obtained using the $3S$ and $2S$ levels

$$\mu_B(m_b) \approx \frac{M[\Upsilon(3S)] - M[\Upsilon(2S)]}{\epsilon_{30}(\lambda_b) - \epsilon_{20}(\lambda_b)} = 350 \text{ MeV}. \quad (169)$$

For the charmonium Bohr scale we get

$$\mu_B(m_c) \approx \frac{M[\psi(2S)] - M[J/\psi(1S)]}{\epsilon_{20}(\lambda_c) - \epsilon_{10}(\lambda_c)} = 319 \text{ MeV}. \quad (170)$$

We can use this value to estimate the mass of the $\psi(3S)$ state as

$$\begin{aligned} M[\psi(3S)] &\approx M[\psi(2S)] + \mu_B(m_c)(\epsilon_{30}(\lambda_c) - \epsilon_{20}(\lambda_c)) \\ &= 4041 \text{ MeV}. \end{aligned} \quad (171)$$

We expect order α_s^4 corrections to these results, meaning corrections of the order of $\mu_f(m_c) = 18.37 \text{ MeV}$. Notice that the only ψ state above the $\psi(2S)$ in Table III which is consistent with these values is the $\psi(4040)$. We conclude that the $\psi(4040)$ is the 3^3S_1 charmonium state. This conclusion is reinforced by the fact that the mass of the 3^3P_2 [the $\chi_{c2}(3930)$] state is below the mass of the 3^3S_1 state and the $\psi(3770)$, $\psi(3823)$, and $\psi(3842)$ assumed as the 3^3D_J charmonium states lie below the S and P waves as required by the inverted spectrum of the Cornell potential.

Once we identify the $\psi(4040)$ as the $\psi(3S)$ charmonium state, we are able to predict the mass of the 3^1S_0 charmonium state [the $\eta_c(3S)$] which according to Eq. (150) must have a mass

$$M[\eta_c(3S)] = M[\psi(3S)] - \mu_f(m_c)f_{30}^{ss}(\lambda_c) = 4003 \pm 4 \text{ MeV}. \quad (172)$$

According to the inverted spectrum obtained with the complete analytical solution of the Cornell potential, the $n = 3$, P -wave states must lie below the $\psi(3S)$ and the $\eta_c(3S)$. This pattern requires the $\chi_{c2}(3930)$ to be the $\chi_{c2}(2P)$ state. The remaining P -waves must lie below the $\chi_{c2}(3930)$ and it is not likely that the $\chi_{c0}(3915)$ be a $\bar{c}c$ state because it is too close to the $\chi_{c2}(3930)$. However, rigorous statements on the identification of the $n = 3$ P -wave charmonium states with the measured $\chi_{c0}(3915)$, $\chi_{c1}(3872)$ and on the missing $h_c(2P)$ requires to do the complete numerical analysis of our main result in Eq. (146) including the complete set of spin-independent operators not considered here.

VI. CONCLUSIONS AND PERSPECTIVES

In this work we completely solve the Cornell potential using the supersymmetric expansion algorithm introduced in Ref. [34]. The solutions are obtained in the form of power series in the normalized string tension λ . It is shown that the coefficients in the series satisfy algebraic recurrence relations which can be solved to the desired order. For these purposes, we wrote a *Mathematica* symbolic code, freely available upon request and calculate the series of the energy levels to order λ^{301} . Although the power series have a small convergence radius, the actual value of the energy functions $E_{nl}(\lambda)$ and of the radial solutions $R_{nl}(r, \lambda)$ can be reconstructed from the power series using Padé approximants. This reconstruction gets more faithful as we increase the number of terms calculated in the power series

and calculations to order λ^{301} allow us to reconstruct the levels up to $n = 4$ with high precision.

The first main result of the complete analytical solution obtained in this work is the prediction of an *inverted spectrum* for the Cornell potential. This means that energy levels depend on two quantum numbers, the principal quantum number n and the orbital quantum number l , this dependence being actually a function of n^2 and $L^2 \equiv l(l+1)$, and for a given n the energy levels *decrease* with increasing l . The radial probabilities have the same shape as the Coulomb ones but the peaks are shifted to smaller radius, thus Cornell eigenstates are more compact than Coulomb eigenstates.

We calculate the critical values of λ for each level, defined by $\epsilon_{nl}(\lambda_{cr}) = 0$ for values of n up to $n = 9$. The energy levels exhibit the crossing phenomena starting with the $n = 4$ level where the $\epsilon_{43}(\lambda)$ crosses with the $\epsilon_{30}(\lambda)$ level for $\lambda \approx 0.4$.

We apply this solution to the problem of the calculation of heavy quarkonium properties. The heavy quarkonium masses depend on two well-defined energy scales, the perturbative physics scale $\mu_Q = 2m_Q$, the natural scale for the Bohr-like levels $\mu_B = 4m_Q\alpha_s^2/9$, and on the normalized energy solutions, $\epsilon_{nl}(\lambda)$. Collecting the masses of all the heavy quarkonium states considered as “well-established” by the Particle Data Group we show that the so far measured heavy quarkonium masses exhibit the inverse spectrum pattern predicted by the Cornell potential.

A first estimate of the values of the parameters $\bar{\lambda}_Q$, $\bar{m}_Q$, $\bar{\alpha}_s(\bar{m}_Q)$ is extracted considering that the Bohr-like levels correspond to the average values of the physical quarkonium masses for both $Q = b, c$ and using the experimental values for the lowest lying $\bar{M}_{10}$, $\bar{M}_{20}$, and $\bar{M}_{21}$. Using these values we predict the average masses for the highest levels. The only level with available data for the complete multiplet is the $n = 3$, $l = 1$ level. The prediction for the corresponding average mass, $\bar{M}_{31}$ agree with the predictions for the central value within 0.2%, which corresponds to 22 MeV.

Details of the structure of heavy quarkonium so far missing like the heavy quarkonium radius and the heavy quark squared velocities can be assessed from our solution. We find striking differences in these observables with respect to the Coulomb values. Indeed, a calculation of the inverse mean radius yields a result $\langle nl|(r/a_Q)^{-1}|nl\rangle$ which differ for its Coulomb $1/n^2$ behavior. The non-perturbative interactions mimicked by the linear term yield smaller radius than the Coulomb interaction for the n levels, which in a given level n increases with the value of the angular momentum l . As to the average squared velocity we find $\langle nl|v^2|nl\rangle = C_{nl}(\lambda)\alpha_s^2$ as expected from NRQCD, but the proportionality constant is large compared with the Coulomb value $C_{nl}(0) = 16/9n^2$ and the nonperturbative effects causes the squared velocity to depend on the orbital

quantum number l and to have a value around $2\alpha_s^2$ for all levels. This result signals to the relevance of relativistic corrections for all the levels of heavy quarkonium, not only for the ground state.

We consider then the leading relativistic corrections and calculate the corresponding fine structure splittings using Rayleigh-Schrödinger perturbation theory. The second main result of this calculation is the expression of the masses of the $n^{2S+1}L_J$ heavy quarkonium states as an expansion in $\alpha_s^2(m_Q)$. Indeed, the formal m_Q^{-2} suppression of the relativistic corrections is actually canceled by m_Q^2 factor from the involved matrix elements (which include the nonperturbative effects of the linear term) leaving actually a power series in α_s^2 . The spectrum involves now three well-defined energy scales; the perturbative scale given by $\mu_Q(m_Q) = 2m_Q$, the scale dictating the Bohr-like levels, $\mu_B = \frac{4}{9}m_Q\alpha_s^2(m_Q)$, and the scale of the fine splittings $\mu_f(m_Q) = \frac{64}{243}m_Q\alpha_s^4(m_Q)$. The masses of heavy quarkonium states are given in terms of these scales, the normalized energy solutions $\epsilon_{nl}(\lambda)$ and the average values of powers of the heavy quarkonium radius normalized to the Bohr radius a_Q which are dimensionless functions $f^{(k)}(\lambda)$ depending only on the normalized string tension λ .

The first general prediction arising from this calculation is that the n^3S_1 states are heavier than the n^1S_0 states. Also, we conclude that the masses of the 3P_J -wave states have a well-defined ordering given by $M[n^3P_2] > M[n^3P_1] > M[n^3P_0]$ for $n = 2, 3$. This ordering is clearly exhibited by the measured states in the $n = 2, 3$ levels of bottomonium and by the measured states in $n = 2$ level of charmonium. Similarly, the masses of the 3D_J states of heavy quarkonium are predicted to satisfy the following hierarchy: $M[n^3D_3] > M[n^3D_2] > M[n^3D_1]$ for $n = 2, 3$.

A confident extraction of the physical value of the normalized string tension λ_Q for heavy quarkonium is done from data on the fine splittings of the lowest lying $l = 0$ states, finding $\lambda_b = 0.361 \pm 0.054$ and $\lambda_c = 0.482 \pm 0.033$. These values and the obtained solutions allow us to fix the fine splitting heavy quarkonium scales to $\mu_f(m_b) = 10.86 \pm 0.54$ MeV and $\mu_f(m_c) = 18.37 \pm 0.36$ MeV. With these values and the obtained normalized probabilities evaluated at the origin, we are able to predict the values of the masses of the $\eta_b(3S)$ and $\eta_b(4S)$ states.

An estimate of the Bohr scales from data of the $n = 1, 2$ S -wave states yields $\mu_B(m_b) \approx 343$ MeV and $\mu_B(m_c) \approx 319$ MeV. These results allow us to identify the $\psi(4040)$ state as the 3^3S_1 [the $\psi(3S)$] charmonium state and to conclude that the measured masses of the $\psi(3842)$, $\psi(3823)$, and $\psi(3770)$ are consistent with the identification of these states as the 3^3D_3 , 3^3D_2 , and 3^3D_1 $\bar{c}c$ states, respectively.

In the $n = 3$ P -wave sector we identify the $\chi_{c2}(3930)$ as the 3^3P_2 charmonium state [$\chi_{c2}(2P)$] and conclude that the

remaining P -wave states $h_c(2P), \chi_{c1}(2P), \chi_{c0}(2P)$ must lie in the $[3842, 3925]$ MeV energy range with the P -wave states being heavier than the D -wave states and the 3P_J and 3D_J ordered according to $M[3^3P_2] > M[3^3P_1] > M[3^3P_0] > M[3^3D_3] > M[3^3D_2] > M[3^3D_1]$.

Finally, this paper yields another interesting application of the supersymmetric expansion algorithm to a long standing unsolved potential of primary importance in the phenomenological description of nonperturbative effects in heavy quarkonium physics. The complete solution of this problem allow us to go from the perturbative to the nonperturbative regime in a controlled manner, revealing and quantifying the role of nonperturbative effects in the conformation of heavy quarkonium. There remain many possibilities for the use of our complete analytical solution to the Cornell potential, including the estimate of

fundamental quantities arising in the NRQCD and p NRQCD effective theories and the complete phenomenological analysis of the leading relativistic corrections.

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DATA AVAILABILITY

The data are not publicly available. The data are available from the authors upon reasonable request.

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